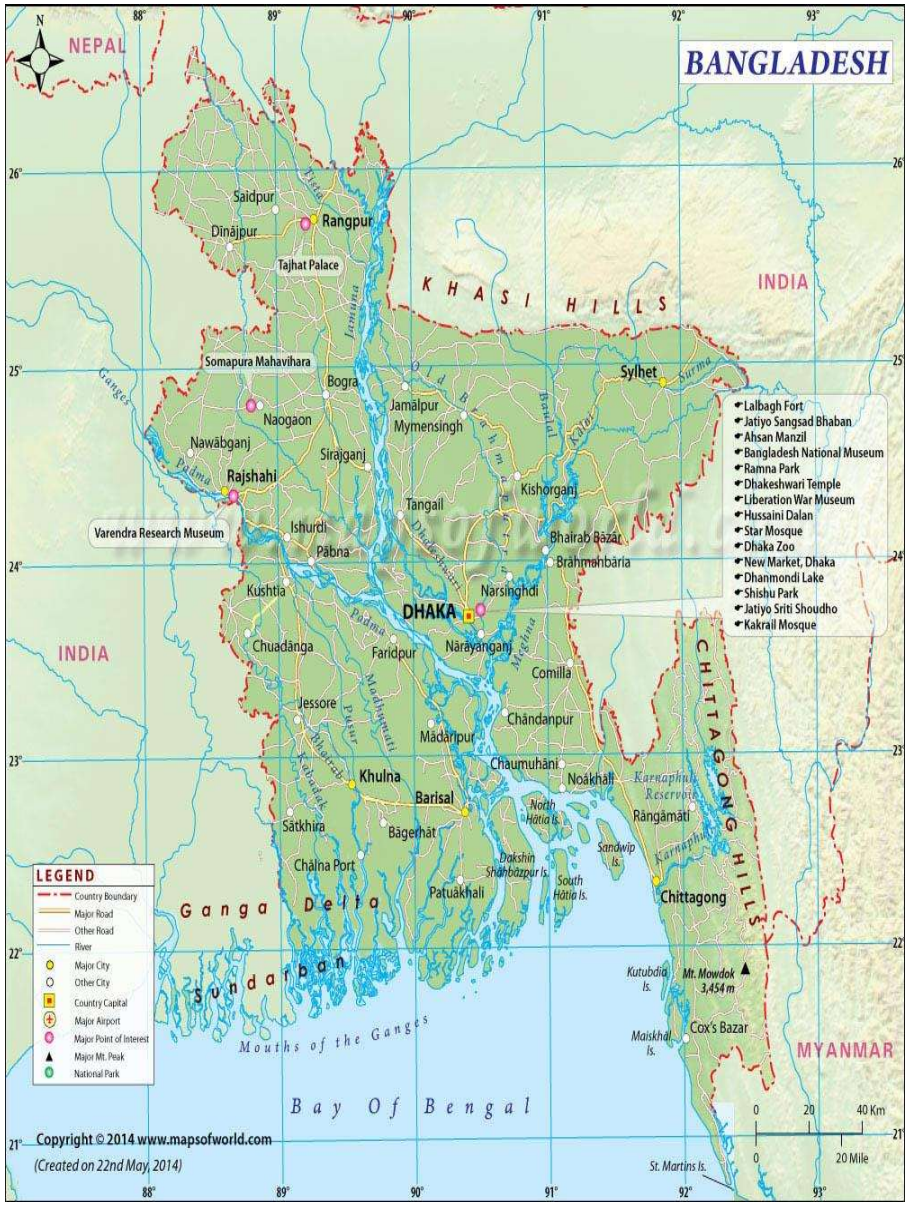
Majority Spanning Trees, Cotrees and Their Applications

Dr. M. Kaykobad
Professor
Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology
Fellow, Bangladesh Academy of Sciences

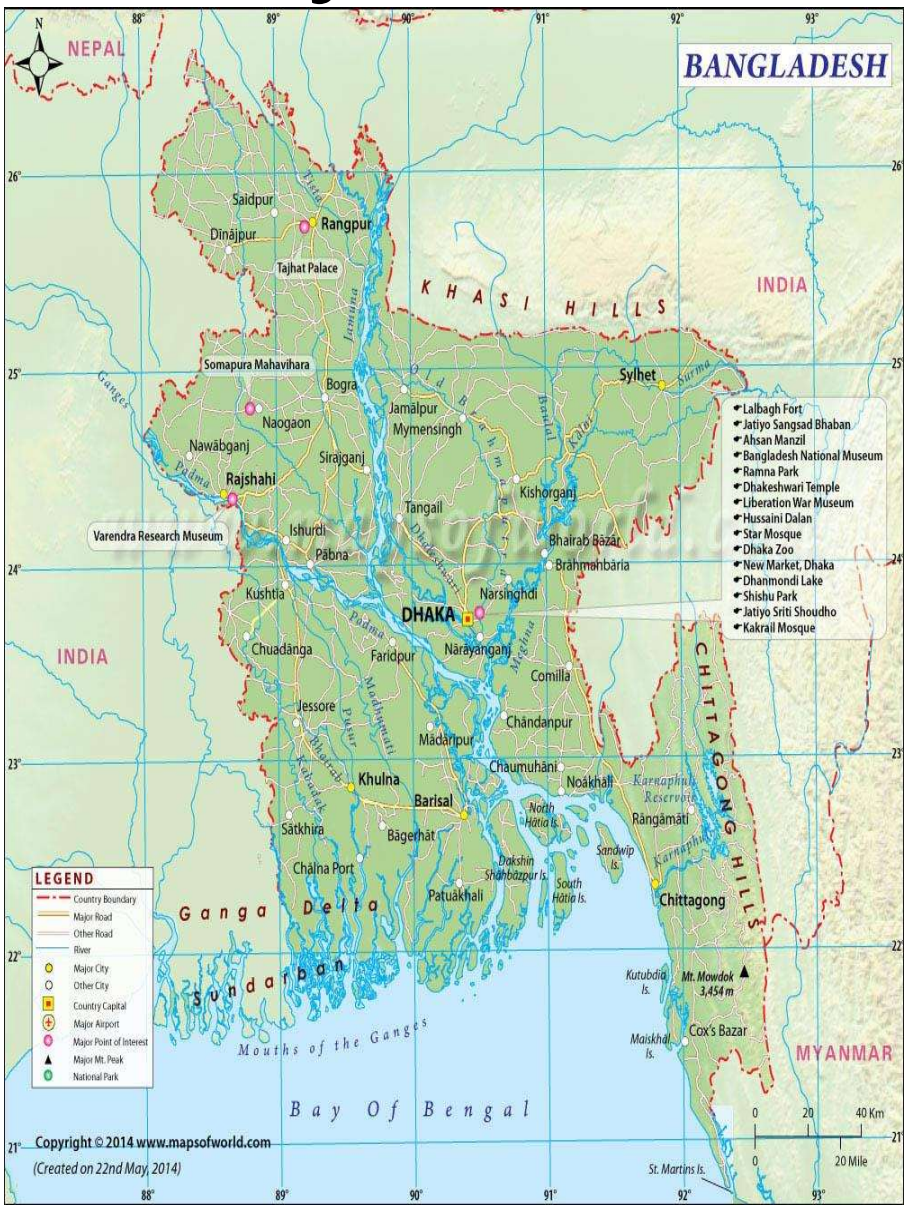
About my country

- Capital and largest city Dhaka (23°42′N 90° 21′E)
- **Ethnic groups** 98% Bengali 2% other
- Religion Islam(86.6%), Hinduism(12.1%), Buddhism (0.6%), Christinity (0.4%)
- Decl of Independence 26 Mar 1971 Liberation of Bangladesh 16 Dec 1971 - Constitution 4 Nov 1972
- Area 147,570 km^2 (92nd) 6.4% Water 56,977 sq mi
- 2016 estimate 162,951,560[6] (8th)• 2011 census 149,772,364[7] (8th)- 1,106/km^2 (2,864.5/sq mi) (10th)USA 35/km^2
- With same population density USA would have resided by 1.6 times the population of earth.
- Friendly country- We are giving shelter to over 1 million refugees from Myanmar even after having such high population density!!!



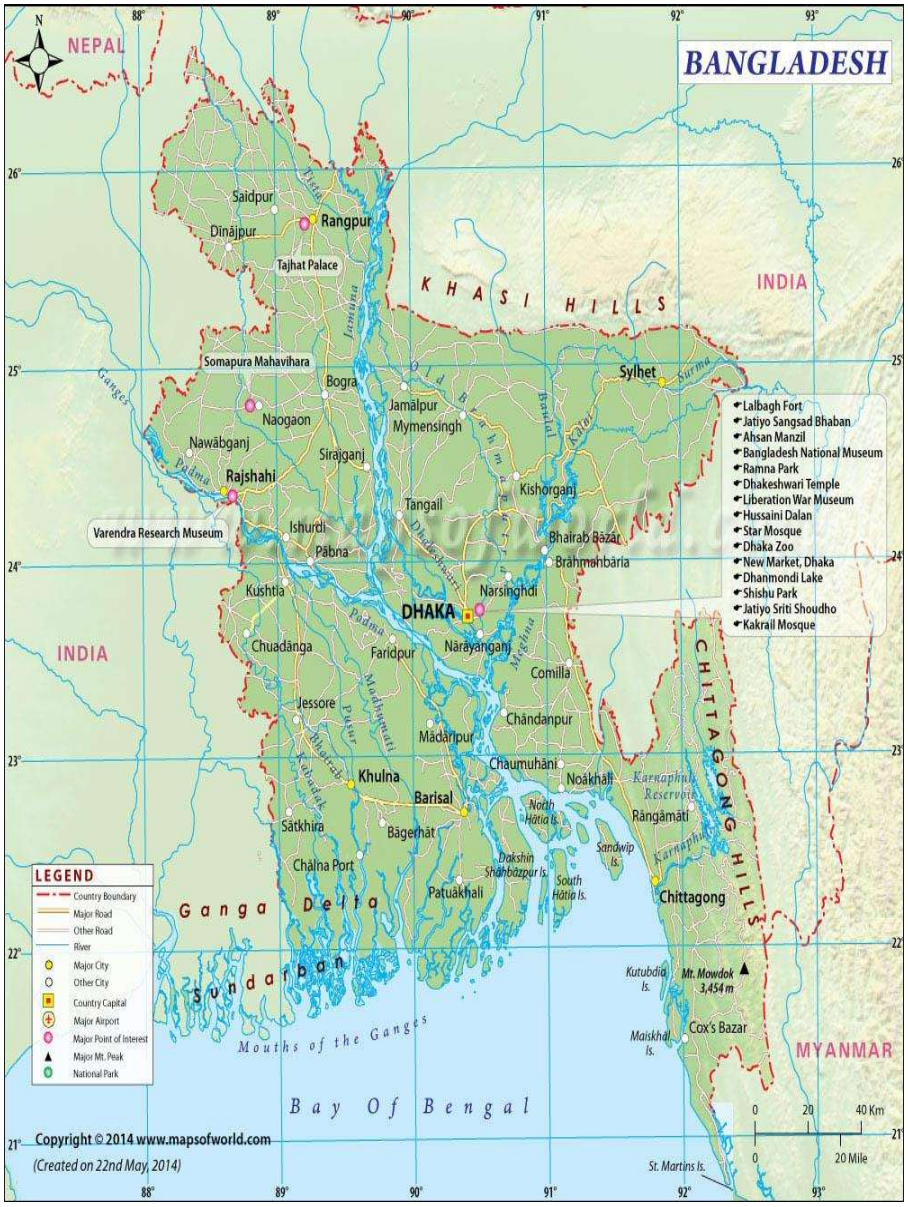
About my country

- GDP (PPP)2017 estimate• Total $686.598 billion[8] (33rd)• Per capita $4,207[8] (139th) GDP PPP 2015 est - $572.440 billion(34th) - Per capita $3,581[7] (144th) Per capita$3,581 (144th)
- GDP (nominal)2017 estimate• Total $248.853 billion[8] (45th)• Per capita $1,524[8] (148th) Gini (2010)32.1[9]
- HDI (2016) 0.579[10] medium · 139th Currency Taka BDT
- Time zone BST (UTC+6)
- Date format dd-mm-yyyy
- Drives on the left



About my country

- Notable personalities- Father of the nation Bangabandhu Sheikh Mujibur Rahman, Rabindra Nath Tagore(NL 1913), Amartya Sen (NL, 1998), Muhammad Yunus(NL, 2006), Satyen Bose, JC Bose, Meghnad Saha, FR Khan
- Primary Level (years 1 to 5),
- Junior Level (years 6 to 8)
- Secondary Level (years 9 to 10)
- Higher Secondary Level (years 11 and 12)
- Tertiary Level



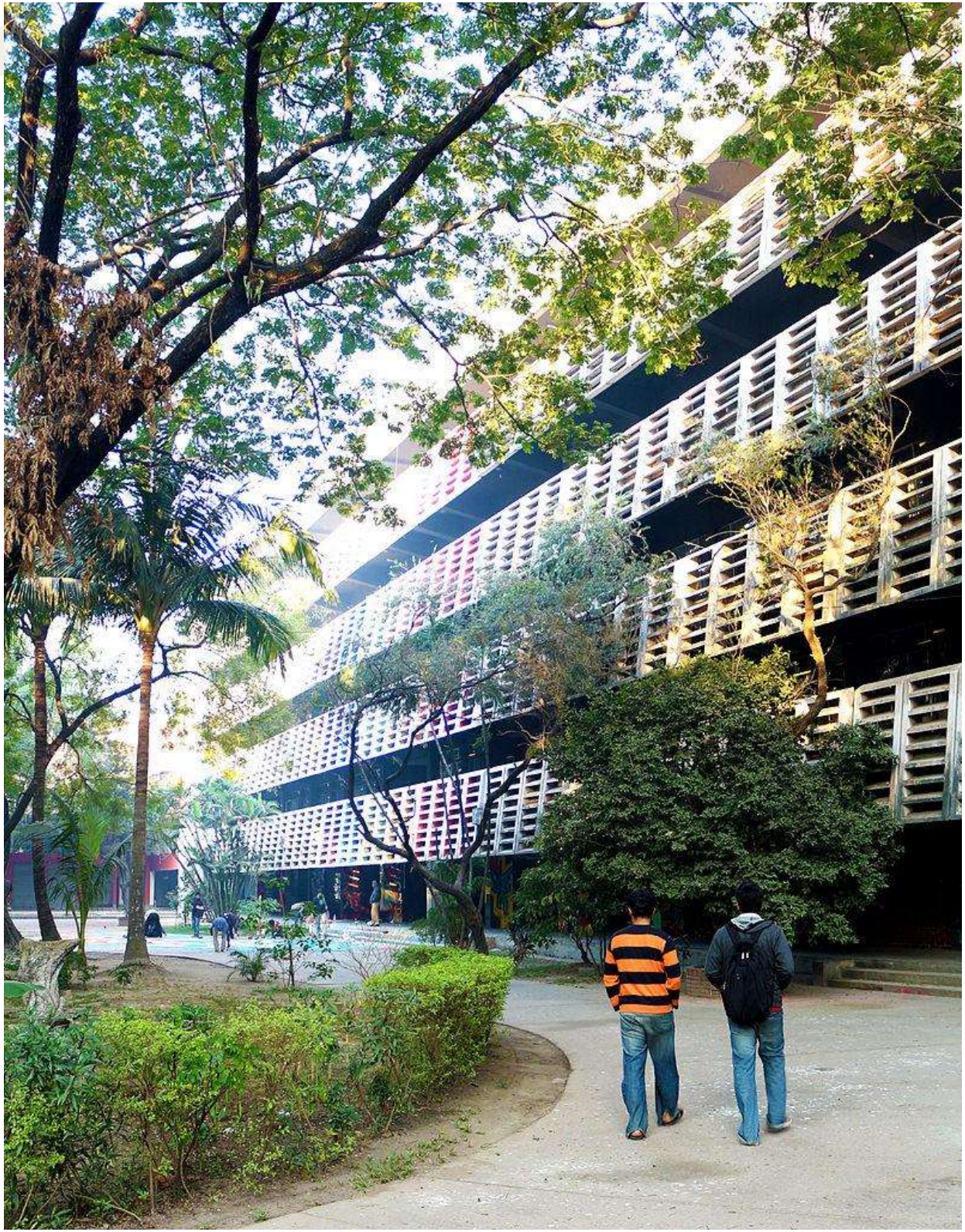
Bangladesh University of Engineering and Technology
বাংলাদেশ প্রকৌশল বিশ্ববিদ্যালয়

Established 1876: Dhaka Survey School 1912: Ahsanullah School of Engineering, 1947: Faculty of Engineering, University of Dhaka 1962: EPUET
1971: BUET Academic affiliations UGCEndowment৳ 1.024 billion (2015-16) ($ 13 million) Chancellor President of Bangladesh Vice-Chancellor Saiful Islam Academic staff 660 [Administrative staff 1,566 Students10,644 (June 015) Undergraduates 5,206 (June 2015) Postgraduates 4,805 (June 015) Doctoral students 440 (June 2015) Other students 193 (June 2015) Location Dhaka, Bangladesh Campus Urban, 91.37 acres (0.37 km^2) [1]Nickname BUET Websitebuet.ac.bd

Bangladesh University of Engineering and Technology







BUET Faculties and Departments

Faculties and Departments Academic activities are undertaken by eighteen departments under five faculties.Twelve departments offer undergraduate courses. All of the departments aside from the department of Humanities offer postgraduate courses.

Faculty of Architecture
Department of Architecture (Arch)
Department of Urban and Regional Planning (URP)
Department of Humanities (Hum)
Faculty of Civil Engineering
Department of Civil Engineering (CE)
Department of Water Resources Engineering (WRE)
Faculty of Electrical and Electronic Engineering
Department of Electrical and Electronic Engineering (EEE)
Department of Computer Science and Engineering (CSE)
Department of Biomedical Engineering (BME)
Faculty of Mechanical Engineering
Department of Mechanical Engineering (ME), Industrial and Production Engineering(IPE), Dept of Naval Architecture and Marine Engineering(NAME)

BUET Faculties and Departments

Faculty of Engineering
Department of Chemical Engineering (ChE)
Department of Materials and Metallurgical Engineering (MME)
Department of Chemistry (Chem)
Department of Mathematics (Math)
Department of Physics (Phys)
Department of Petroleum and Mineral Resources Engineering (PMRE)
Department of Glass and Ceramic Engineering (GCE)

Journals and Research Bulletins
Journal of Mechanical Engineering Research and Developments
Chemical Engineering Research Bulletin
Bangladesh Journal of Water Resources Research
Electrical and Electronic Engineering Research Bulletin
Protibesh (Research Journal on Architecture)

BUET Student Organizations

Murchhona : BUET (Cultural Organization)
Kantho:BUET (Recitation Club)
BUET Nuclear Engineering Club (BNEC) https://bnec.club
Badhan (Blood donation), BUET Zone
BUET Robotics Society (BRS) http://buetroboticssociety.org/
BUET Energy Club (BEC)
BUET Career Club (BCC)
BUET Debating Club (BUETDC)
BUET Drama Society
BUET Film Society
BUET Photographic Society (BUETPS)
House of Volunteers : BUET
Satyen Bose Club (a Science organization named after Satyendra Nath Bose)
BUET Rover Group
BUET Literary Society
Civil Engineering Students' Association (CESA)
Water Resources Engineering Students' Association (WRESA)

BUET Institutes

At present there are six institutes in BUET. These institutes offer postgraduate diplomas, master's and doctoral degrees.

Institute of Water and Flood Management (IWFM)
Institute of Appropriate Technology (IAT)
Institute of Information and Communication Technology (IICT)
Accident Research Institute (ARI)
BUET-Japan Institute of Disaster Prevention and Urban Safety (BUET-JIDPUS)
Institute of Nuclear Power Engineering (INPE)

Research Centers and Directorates

Research Centers
Centre for Energy Studies (CES)
Centre for Environmental and Resource Management (CERM)
Biomedical Engineering Centre (BEC)
Bureau of Research, Testing and Consultation (BRTC)
International Training Network Centre (ITN)
Bangladesh Network Office for Urban Safety (BNUS)
Directorates
Directorate of Advisory, Extension and Research Services (DAERS)
Directorate of Students' Welfare (DSW)
Directorate of Planning and Development (P&D)
Directorate of Continuing Education (DCE)

Research Centers and Directorates

Directorates
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Our Notable Alumni

Fazlur Rahman Khan, a Bangladeshi-American structural engineer and architect, considered the "father of tubular designs for high-rises", recipient of the Independence Day Award in the year 1999
, A.K.M. Fazle Hussain, academic at the University of Houston,[39] has been awarded the highest awards of the American Physical Society, American Society of Mechanical Engineers and American Institute of Aeronautics and Astronautics
Professor Jamilur Reza Choudhury, National Professor, civil engineer, received Ekushey Podok, Honorary Dr. Eng from Univ of Manchester, Adviser of Care taker Govt, Bangladesh

IEEE Fellows

Professors Waquar Ahmed, Tariq Durrani, Syed A Nasar, Saifur Rahman of Virginia Tech, Dr. Md. Azizur Rahman, Ashraful Alam of Purdue, Muhammad H Rashid, Iqbal Hussain, AZM Ekram Hossain, S Alam, Haran Karmaker, M Abul Masroor, M. Fazlur Rahman, Sayeef Salahuddin of Berkeley



Majority Spanning Trees, Cotrees and Their Applications

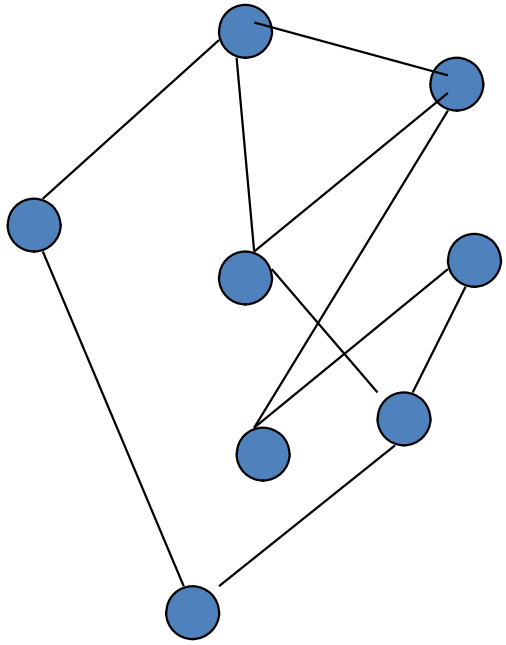
Contents

- Introduction
  - Spanning Tree, Minimum Spanning Tree (mST)
  - mST Algorithms
- Majority Spanning Tree(MST)
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  - Definition of Majority Spanning Trees
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- Theorems
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- Minimum Connection Time(McT) Problem
- Complexity of McT Problem
- Ranking of Round Robin Tournaments
- Complementing one-way roads for equilibrium
- Settling multiple debts efficiently
- Conclusion

Spanning Trees

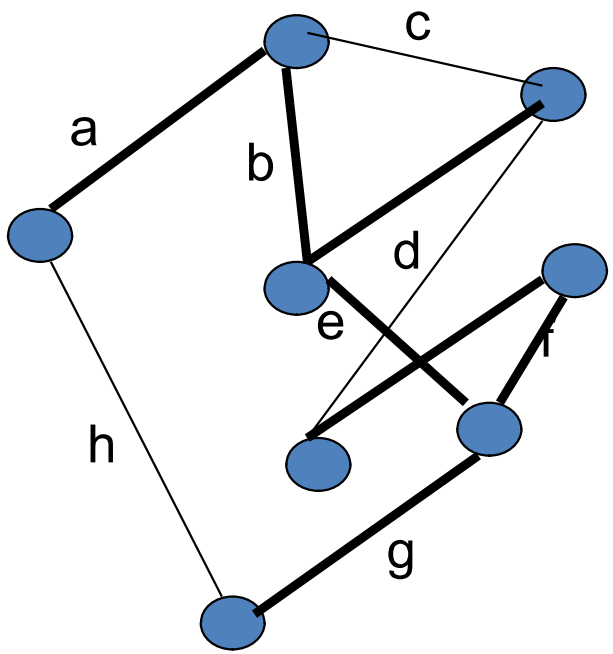
- Let G=(V,E) be a connected graph with vertex set V and edge set E.
- A subgraph that spans (reaches out to) all vertices of a graph is called a **spanning subgraph.**
- A subgraph that is a tree and that spans (reaches out to) all vertices of the original graph is called a **spanning tree.**

- Among all the spanning trees of a weighted and connected graph, the one (possibly more) with the least total weight is called a **minimum spanning tree (mST).**



Algorithms for finding MST

- The well known algorithms are:
  - Kruskal's Algorithm
  - Prim's Algorithm

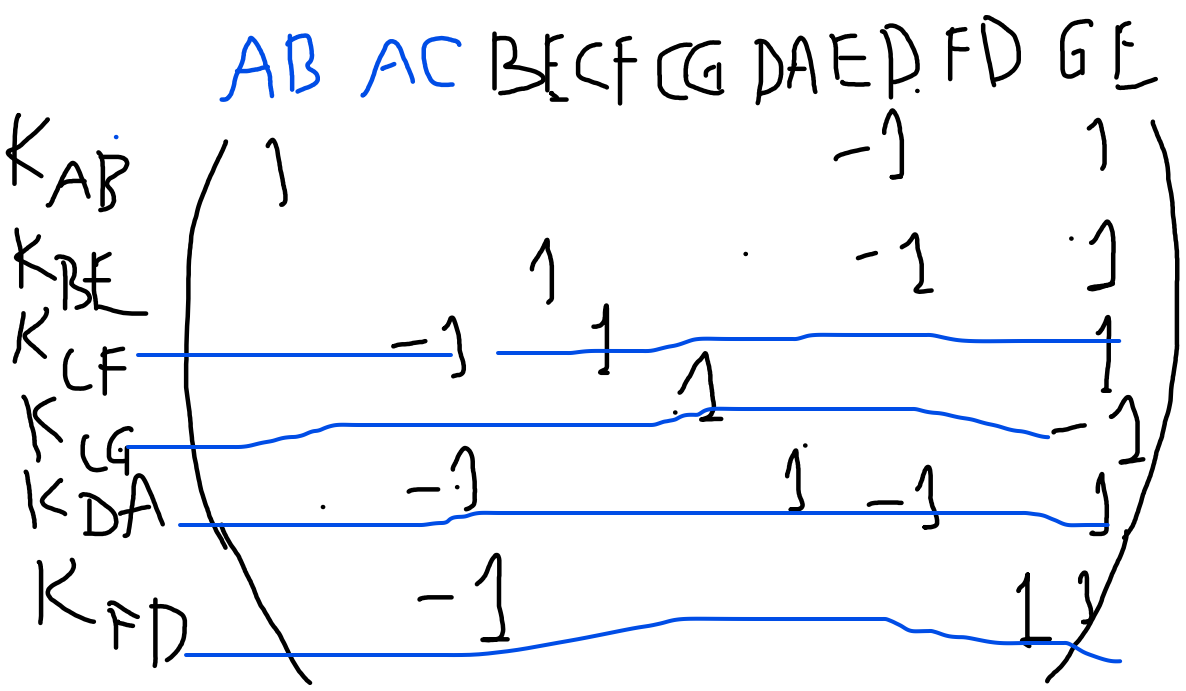


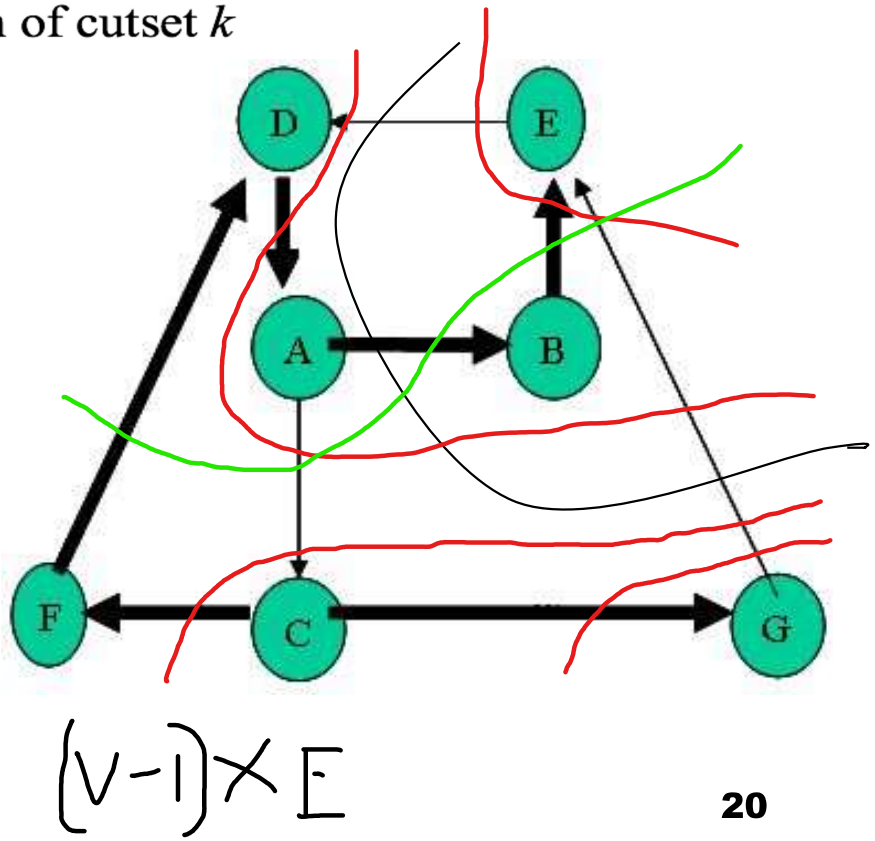
Fundamental Cutsets

- **Fundamental Cutset Arc Incidence Matrix:**

It is defined by- $(|V|-1)\times(|A|-1)$ matrix $Q=(q_{ij})$

$$q_{kj} = \begin{cases} 1, \text{ if edge } j \text{ is in the positive orientation of cutset } k \\ -1, \text{ if the edge } j \text{ is in the negative orientation of cutset } k \\ 0, \text{otherwise} \end{cases}$$





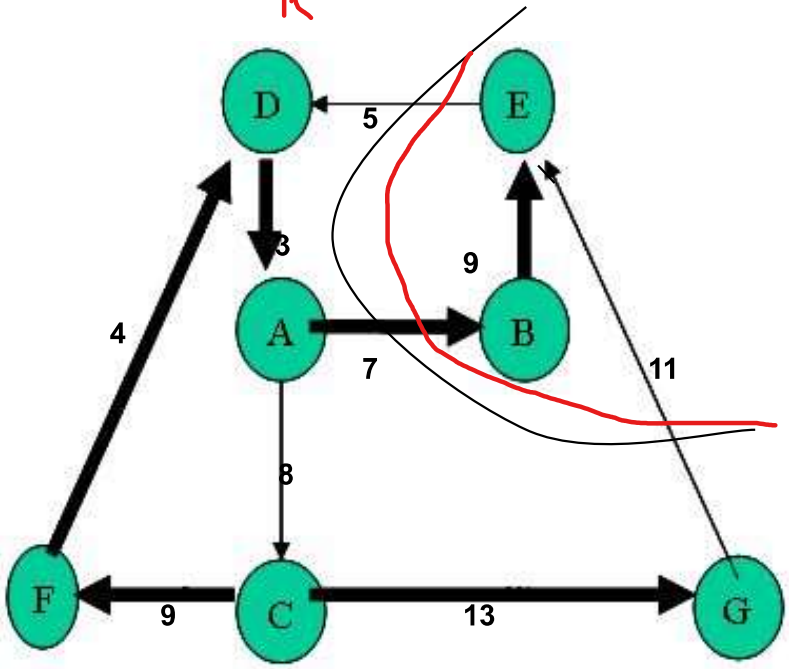
Orientation in Cutsets

- **Fundamental Cutset**
  Let T be any arbitrary spanning tree of G, K_k be the fundamental cutset defined by edge $(i_1,j_1)=e_k \in T$, and

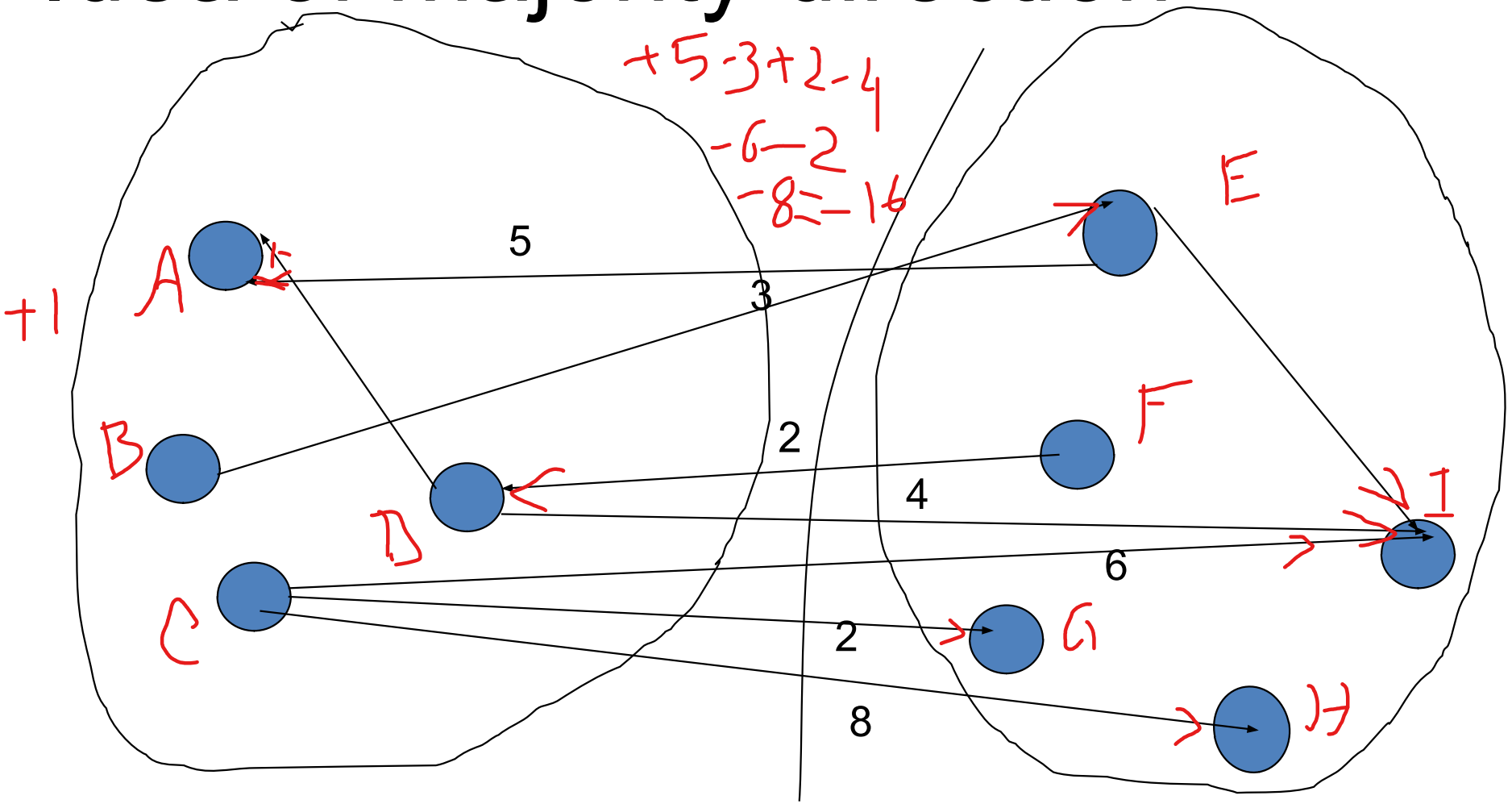
$$r_k = \sum_{(i,j)\in K_k^+} p_{ij} - \sum_{(i,j)\in K_k^-} p_{ij} \qquad (2.1)$$

  where, K_k^+ , K_k^- are the set of forward and reverse edges of the cutset K_k. We define the value of the cutset. K_k by $P(K_k)=r_k$, p_{ij} is the weight of the edge (i,j).

If e_k =(AB) then K_k^+ ={AB, GE}, K_k^- ={ED}
r_k=7+11-5=13



Idea of majority direction



Trips are vertices, edges are connections. One of the connections in the cutset must have 0 waiting time. In particular one in the majority direction. Increasing t_i's by 1 of LHS will reduce waiting time by 3+4+6+2+8 and increase by 5+3+2 hours provided all waiting times are >0

Majority Spanning Trees

- ***Definition:*** *A spanning tree T is said to be a majority spanning tree of the digraph G=(V,E) with real weight function* $P:E \rightarrow R^+$ *if for each fundamental cutset* K_k, *determined by the edges of T,* $P(K_k) \geq 0$

- For example, the value of the cut-set determined by the edge DA is (-5+3-8+11)=1>=0 since DA and GE are in the same orientation whereas AC and ED are in the opposite. In this way cut-set determined by other edges of the spanning tree with thick edges can be shown to have nonnegative weights.

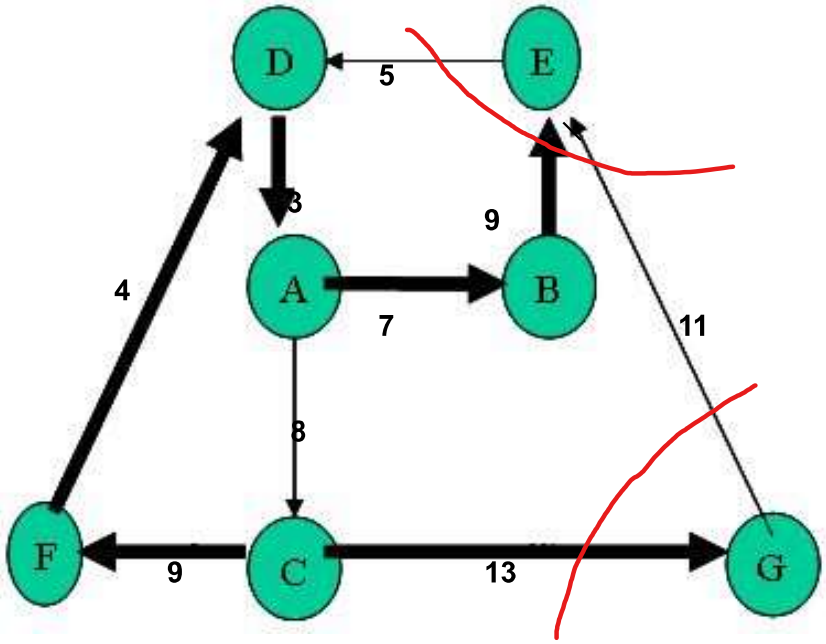


Fig 1: Majority Spanning Tree

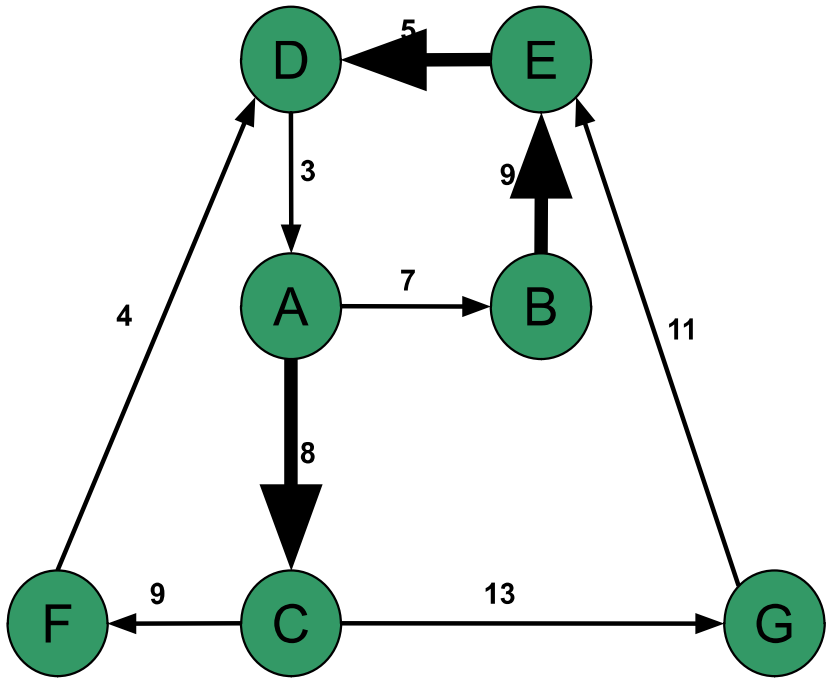
Majority CoTrees

$\overline{T} = G - T$

$C_{AC} = \{AC, CF, FD, DA\}$

Definition: *A cotree* $\overline{T}$ *is said to be a majority cotree of the digraph G=(V,E) with real weight function* $P: E \rightarrow R^+$ *if for each fundamental cycle* C_k*, determined by the edges of T and one edge of* $\overline{T}$, $P(C_k) \geq 0$.

i.e. Let $\overline{T}$ be the cotree consisting of the thick edges as shown in Fig. 2. The fundamental cycles CFDAC, CGEDAC and ABEDA, determined respectively by the edges CF, CG and AB are directed. Evidently, P(CFDAC), P(CGEDAC) and P(ABEDA) are all non-negative, and hence is a majority cotree.

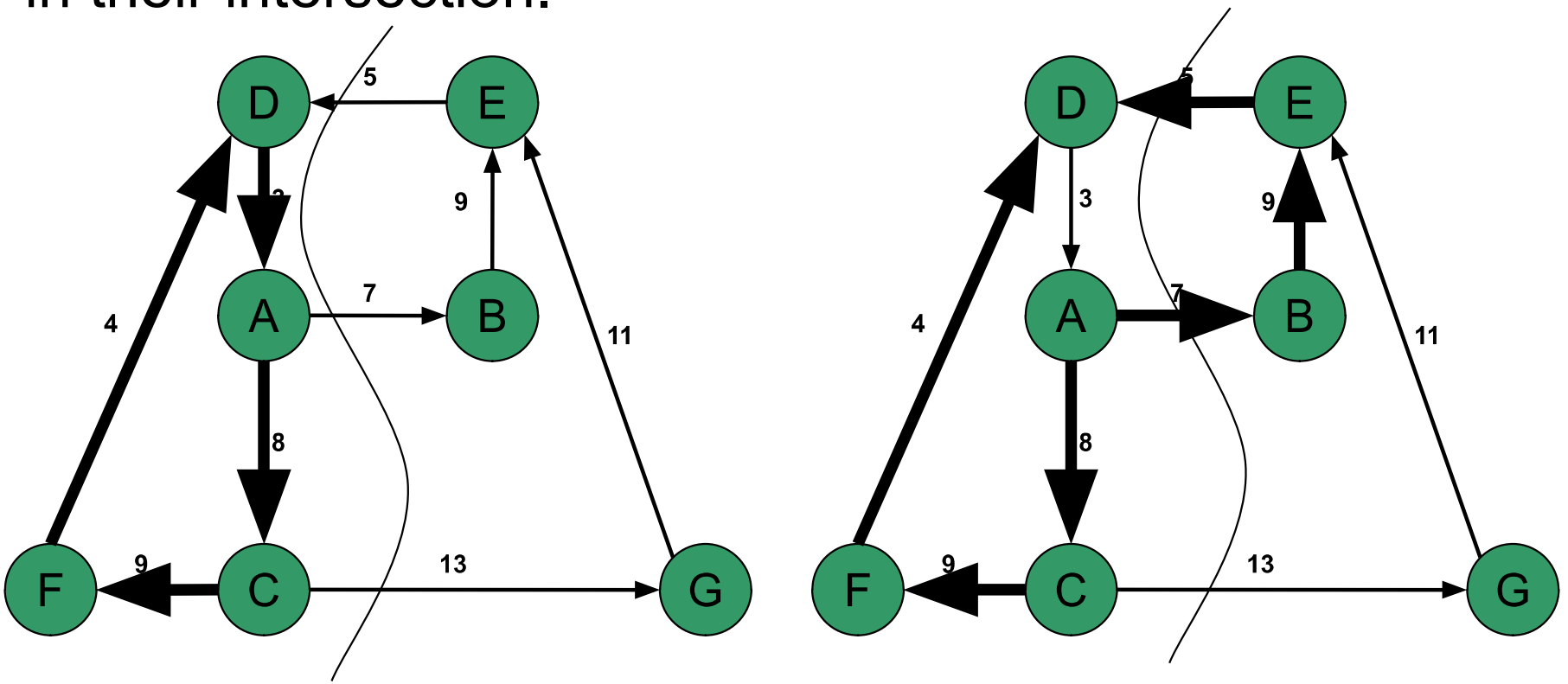


Minimum, Maximum and Majority Spanning Trees

- Each edge of an MinST is minimum (maximum) edge of the cutset defined by itself.
- Each edge of an MajST is in the majority direction of the fundamental cutset it defines.
- Each edge of an MajCT is in the majority direction of the fundamental cycle it defines.
- In this sense they are natural extensions to already existing ideas.

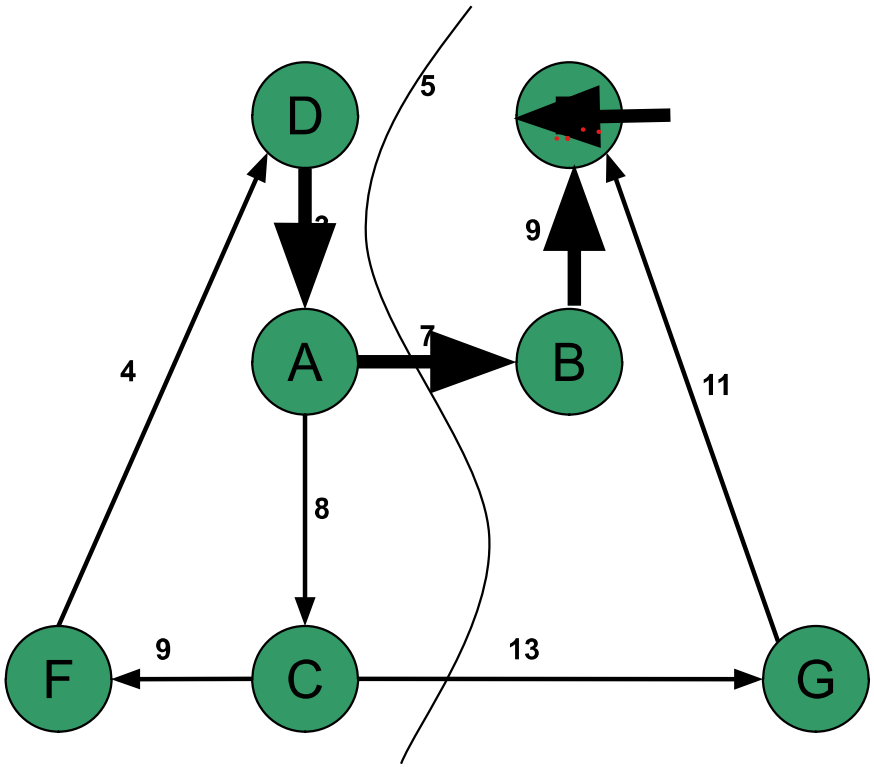
Orientations in cutsets

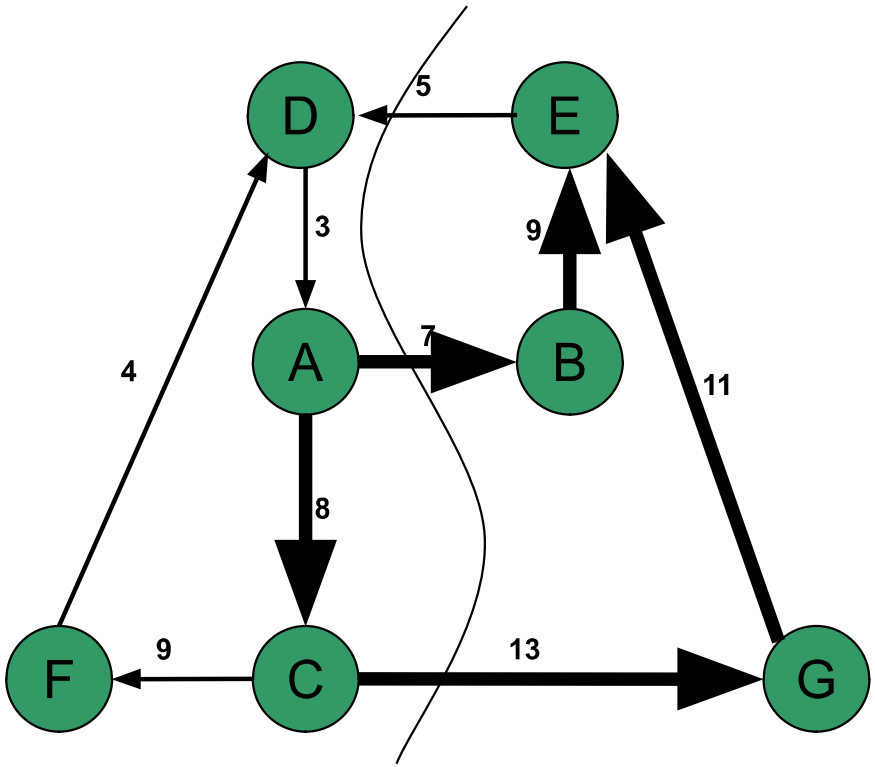
- Let C and K represent sets of edges respectively in a cycle and a cutset. Then there are even number of edges in their intersection.



Orientations in cycles

- If two edges are in the same orientation in a cutset then they are in the opposite orientation in a cycle and vice versa.

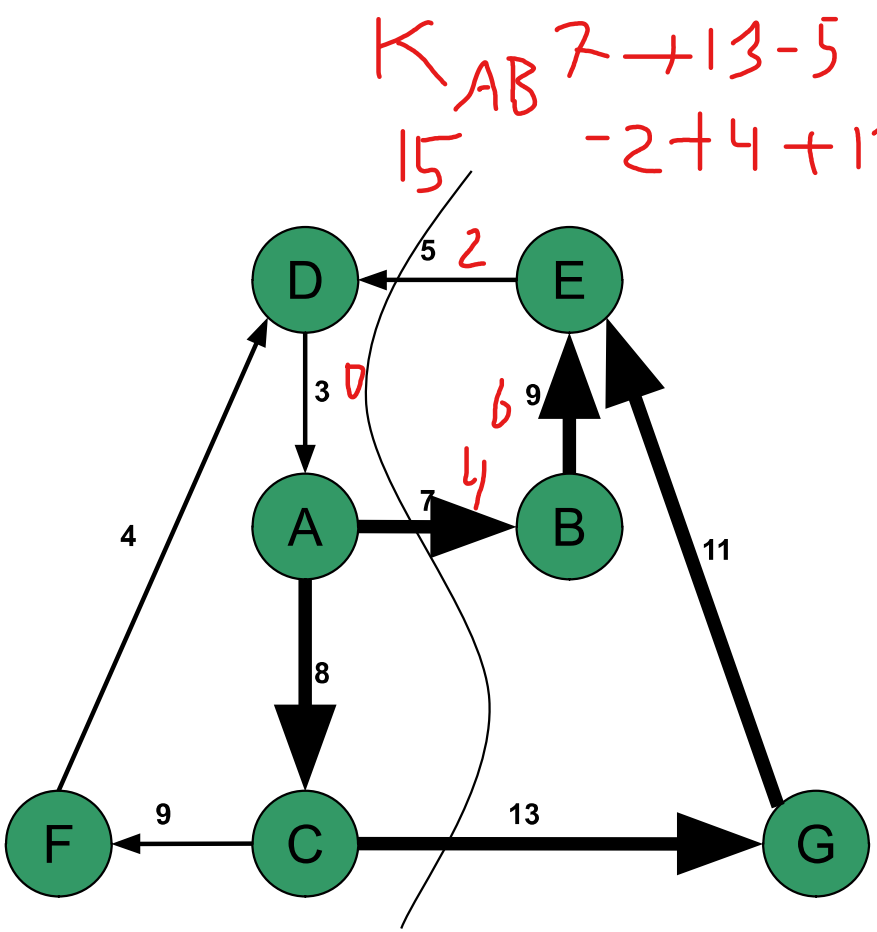


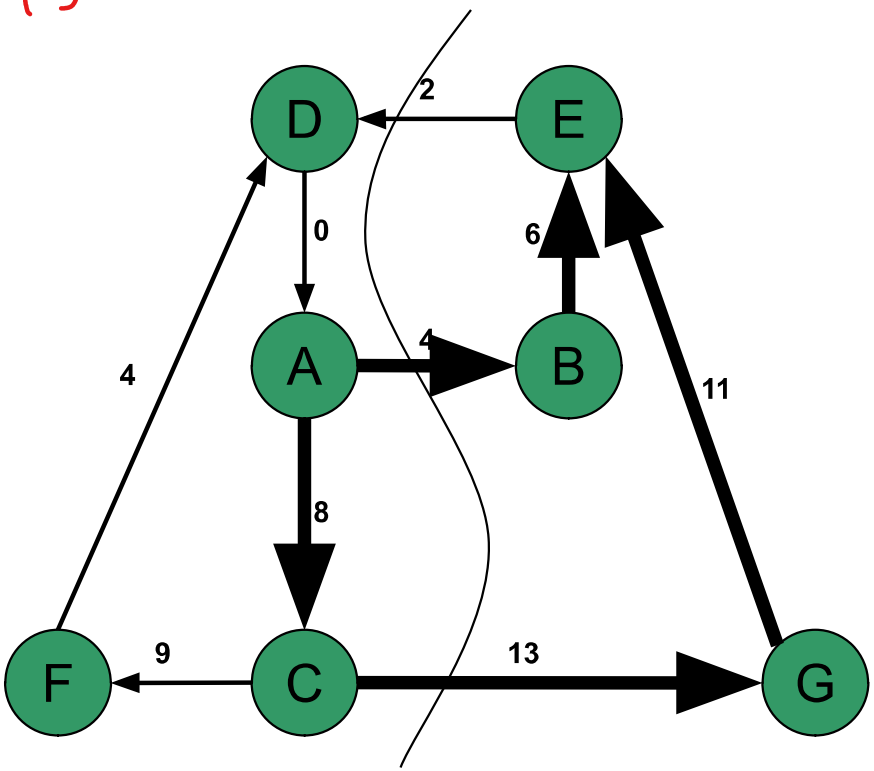


Keeping weights of cutsets/cycles unchanged

- Lemma 1. If weights of edges of a cutset running in the same orientation is reduced by their minimum and weights of edges in the opposite orientation is increased by the same amount, weights of cycles remain unaffected keeping all edge weights non-negative.

Keeping weights of cutsets unchanged

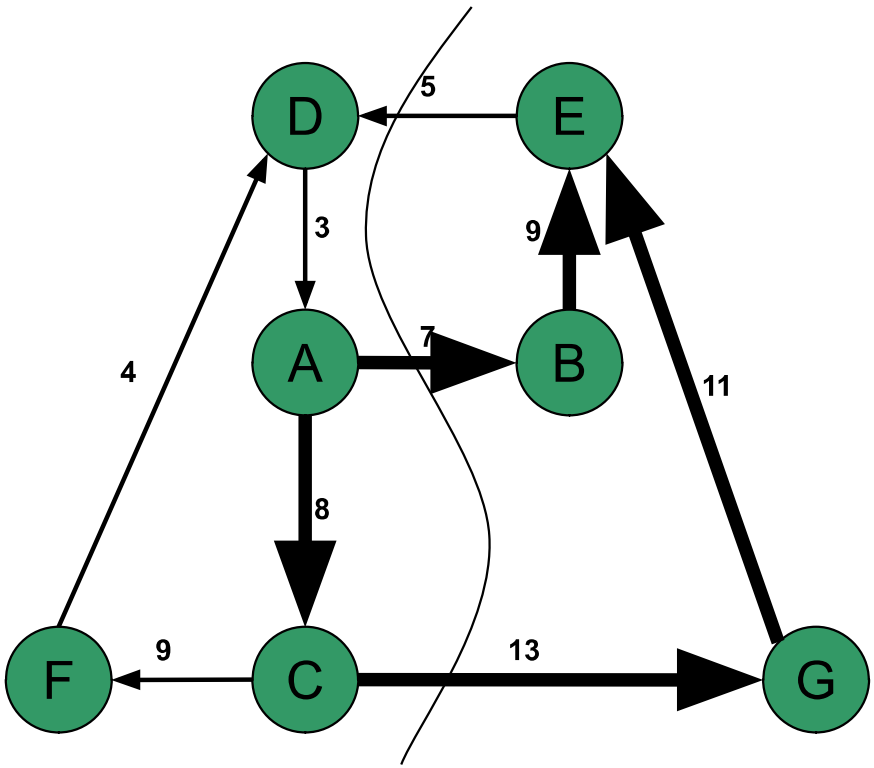


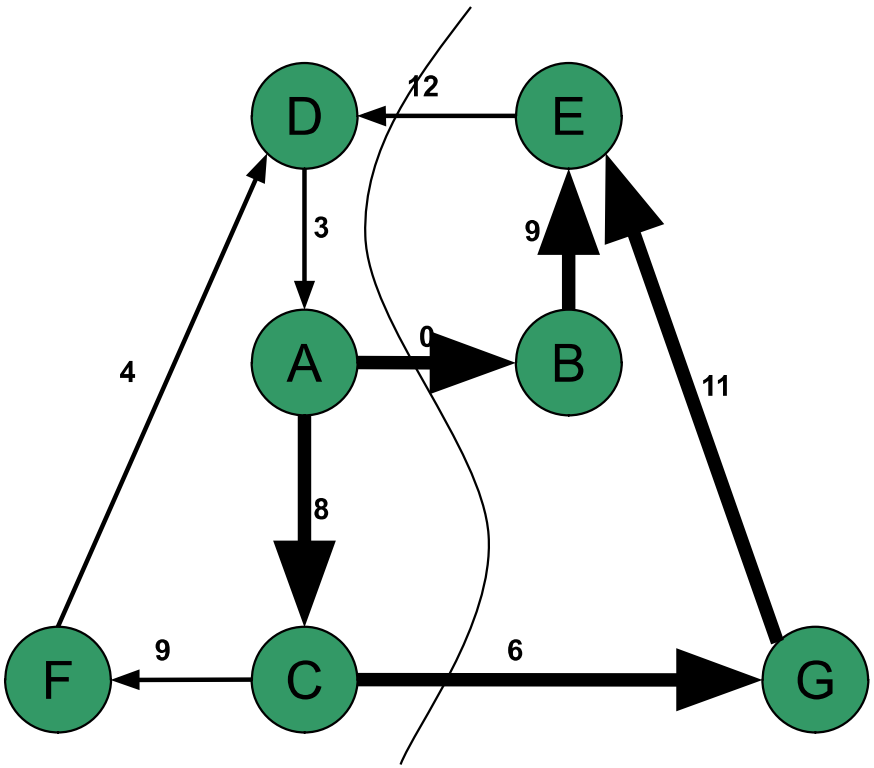


Keeping weights of cutsets/cycles unchanged(continued)

- Lemma 2. If weights of edges of a cycle running in the same orientation of the cycle are reduced by their minimum and weights of edges in the opposite orientation are increased by the same amount, weights of cycles remain unaffected keeping all edge weights non-negative.

Keeping weights of cycles unchanged





Theorems

- **Theorem 2.1:** ***Every digraph G=(V,E) with non-negative weight function P:E → R+ has a majority spanning tree.***
- ***Theorem 2.2:*** ***Every digraph G=(V,E) with non-negative weight function P:E → R+ has a majority cotree.***
- ***Theorem 2.3:*** ***For every digraph G=(V,E) with non-negative weight function P:E → R+ there is a majority spanning tree T such that G-T is a majority cotree.***

Constructive Proof of Theorem 1

- So long as there is a cycle with all its edges having positive weights, apply Lemma 1 to increase the number of 0 edges by at least 1. This will result in a graph in which all positive weighted edges form a spanning forest that can be complemented with 0 edges to form a spanning tree.

Constructive Proof to Theorem 1(contd)

- This must be a majority spanning tree. Since by Lemma 1 changing edge weights in that way does not change weights of any cutset, in particular fundamental ones, whose value is equal to the weight of the spanning tree edge since edges of cotree are 0 weighted.

Constructing a majority spanning tree

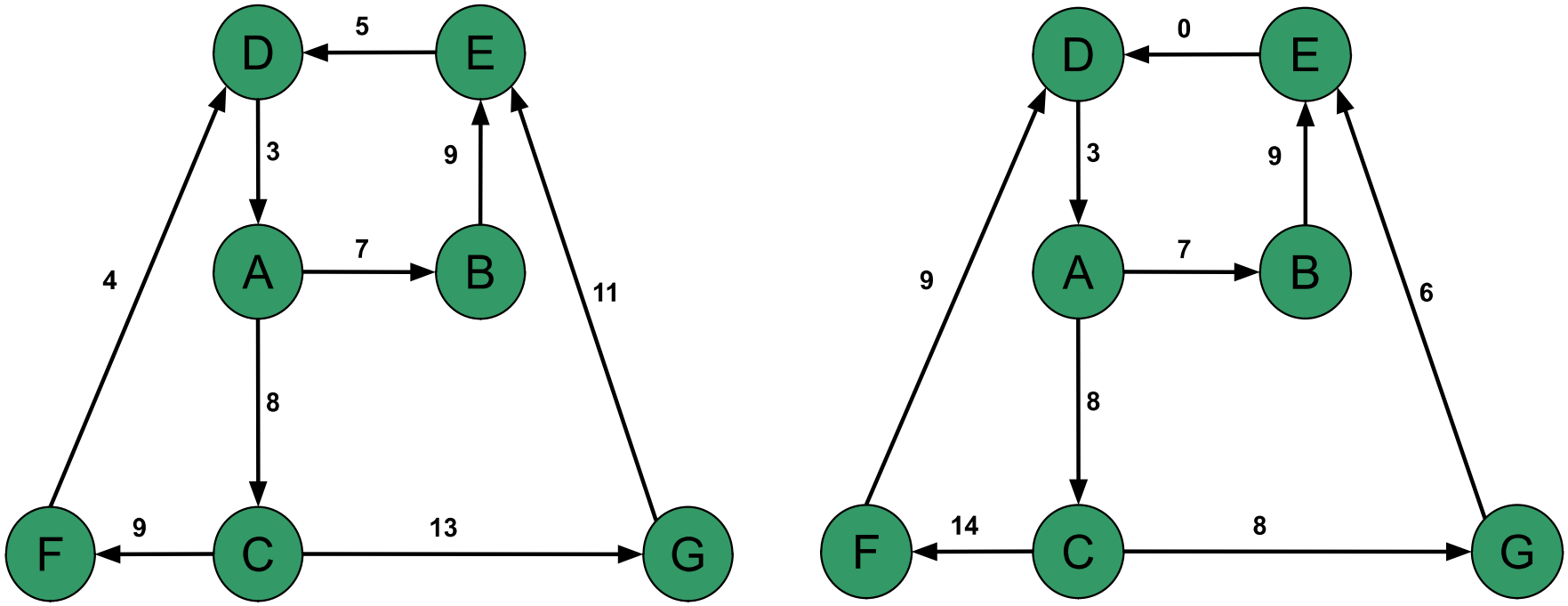


Fig 1a: Constructing MST (change along EDFCGE)

Majority Spanning Trees

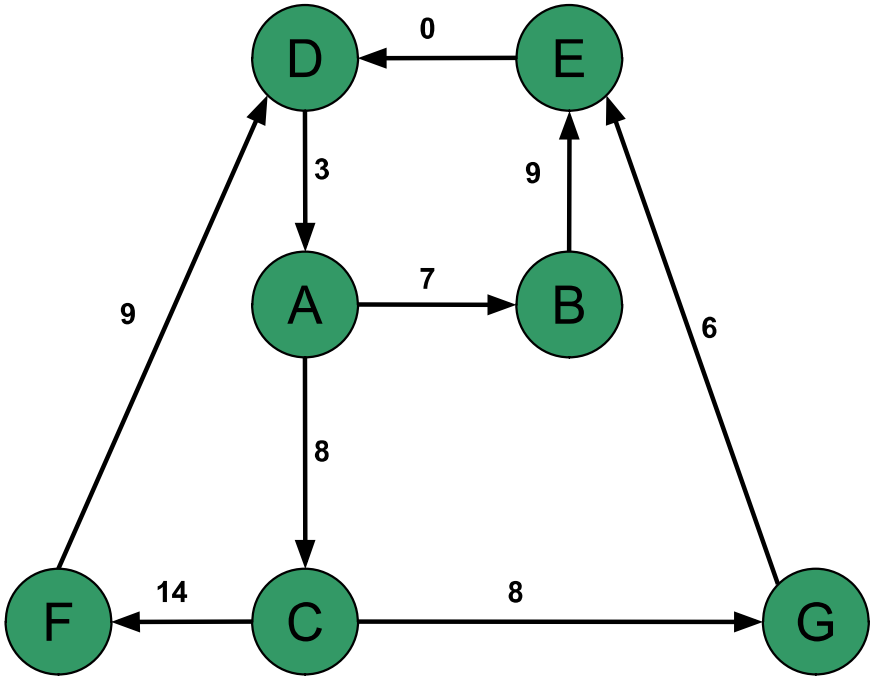


Fig 1a: Constructing MST(GEBACG)

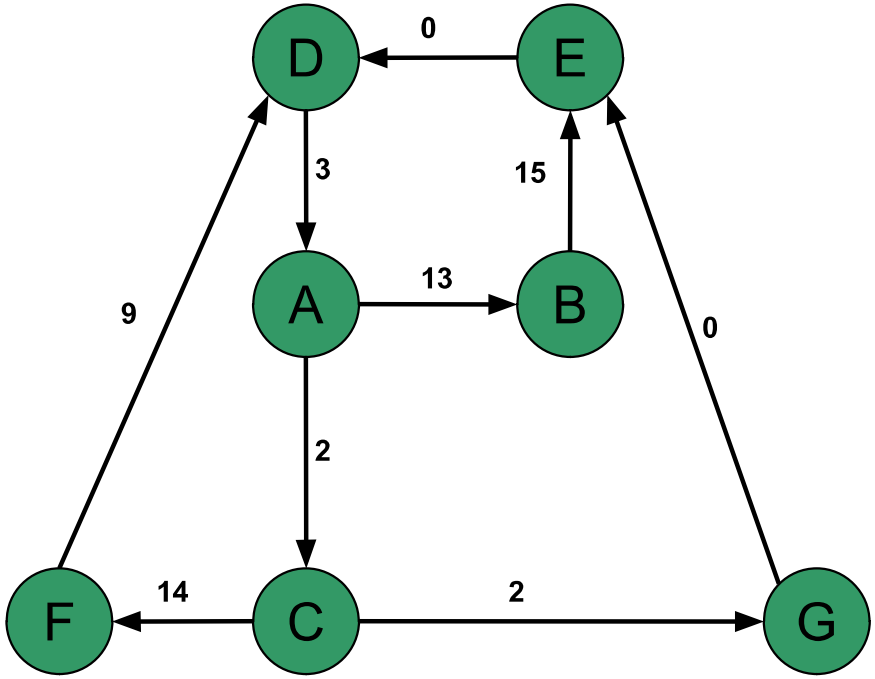


Fig 1b: Constructing MST

Majority Spanning Trees

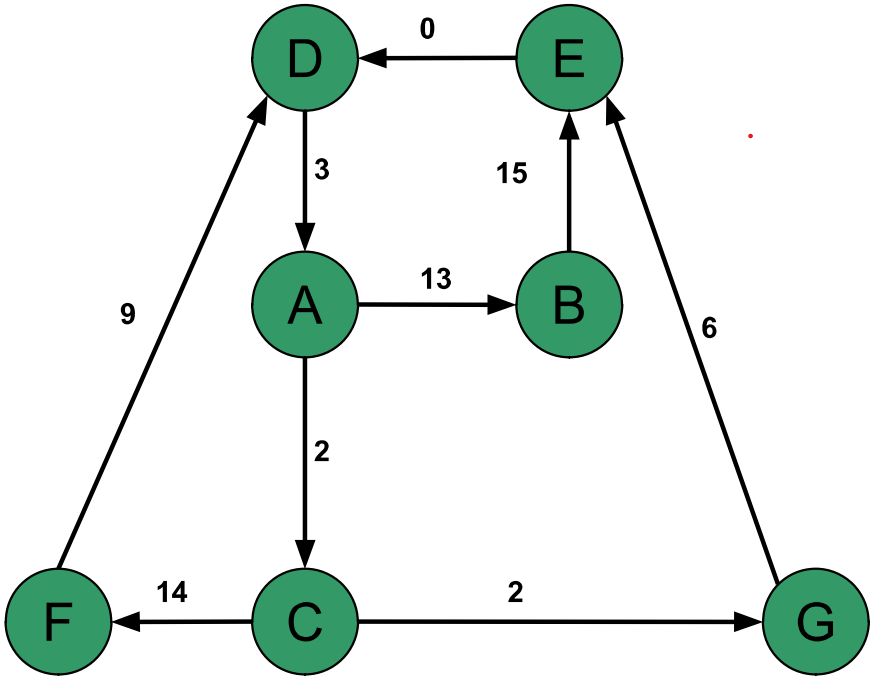


Fig 1b: Constructing MST(ACFDA)

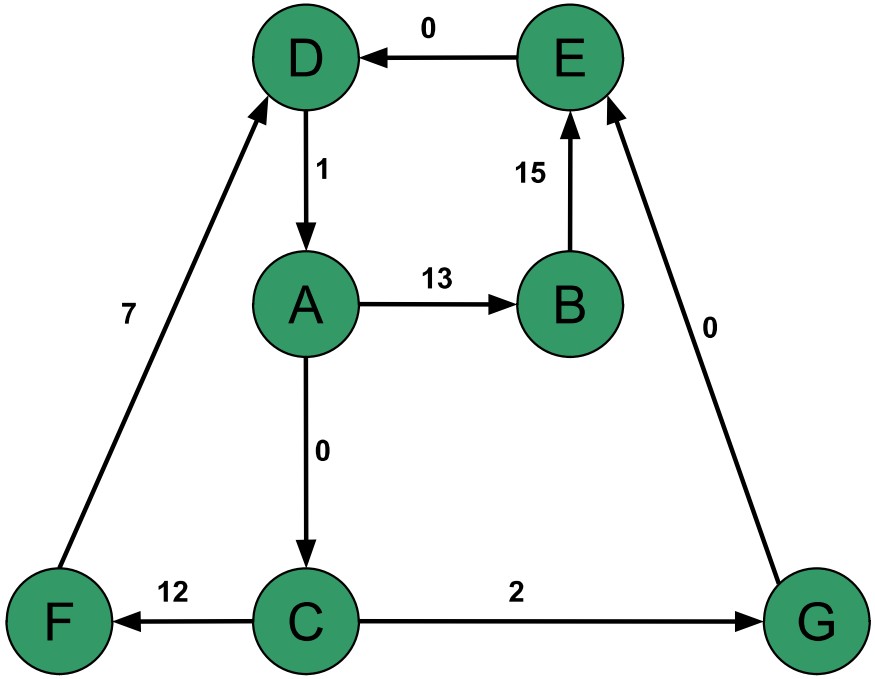


Fig 1c: Constructing MST

Majority Spanning Trees

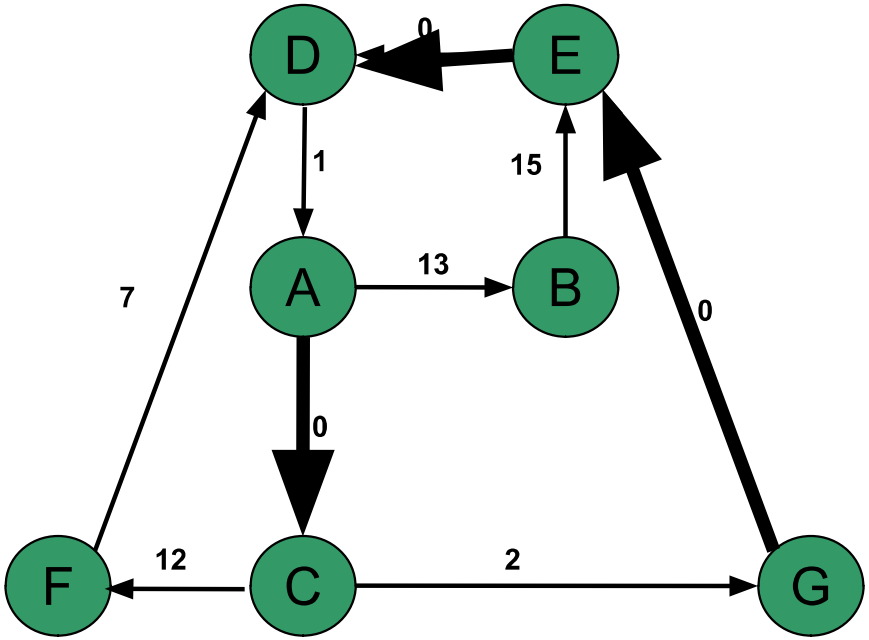
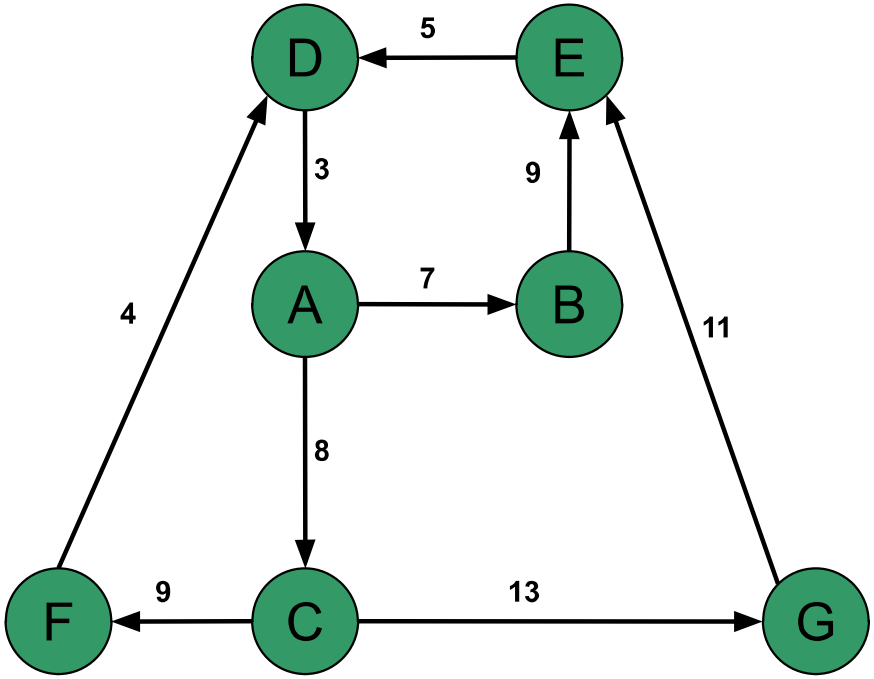


Fig 1c: **Majority Spanning Tree**



Digraph with original weights

Constructive Proof to Theorem 2

- So long as there is a cutset with all its edges having positive weights, apply Lemma 2 to increase the number of 0 edges by at least 1. Now 0 edges will contain a spanning tree. As a result weights of cycles, in particular, fundamental cycles, will be equal to the edge defining the fundamental cycle.

Constructing Majority Cotrees

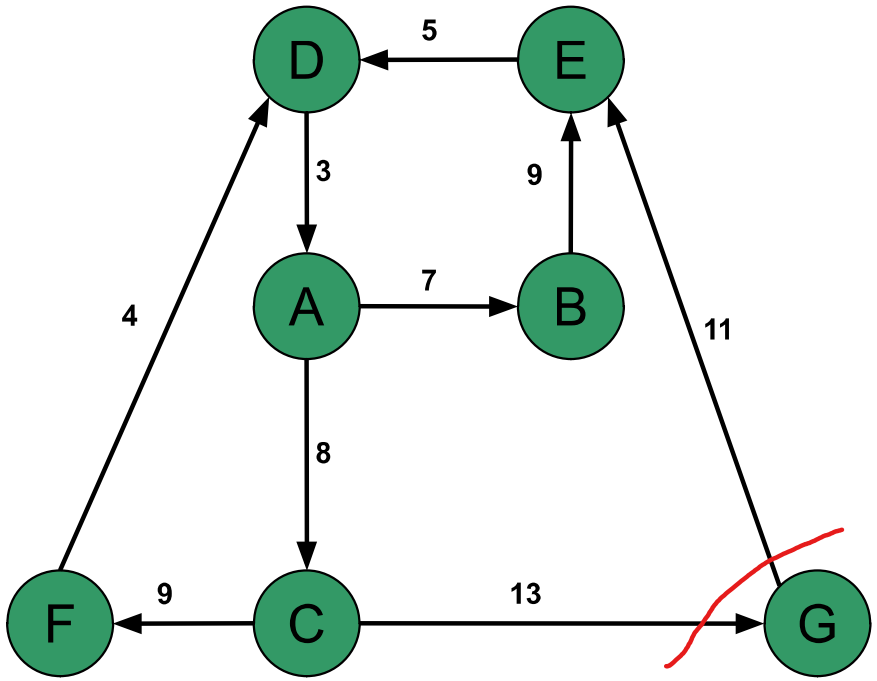


Fig 2a: Constructing a Majority Cotree(G)

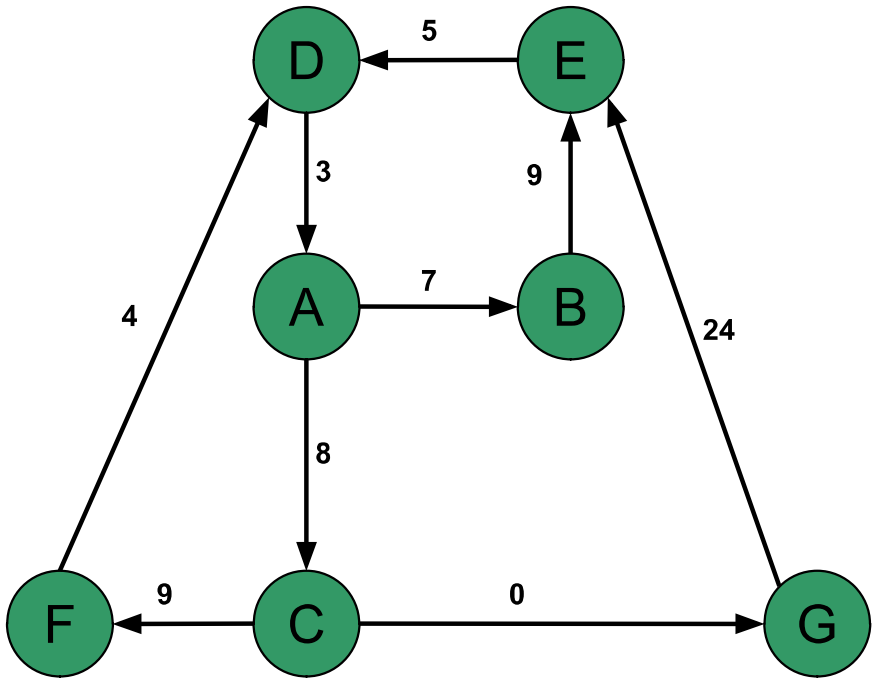


Fig 2b: Constructing a Majority Cotree

Constructing Majority Cotrees

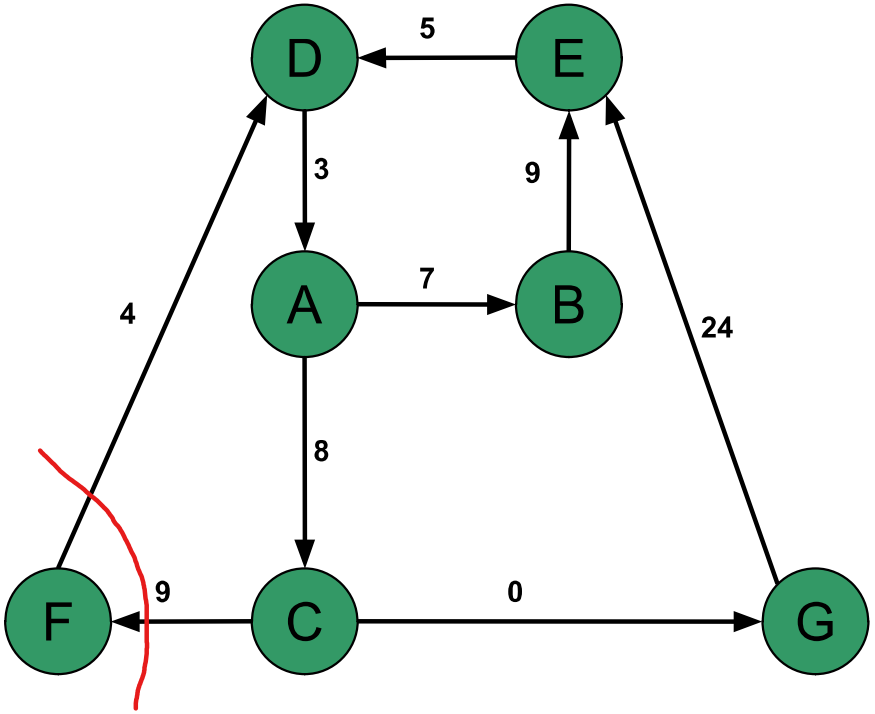


Fig 2b: Constructing a Majority Cotree(F)

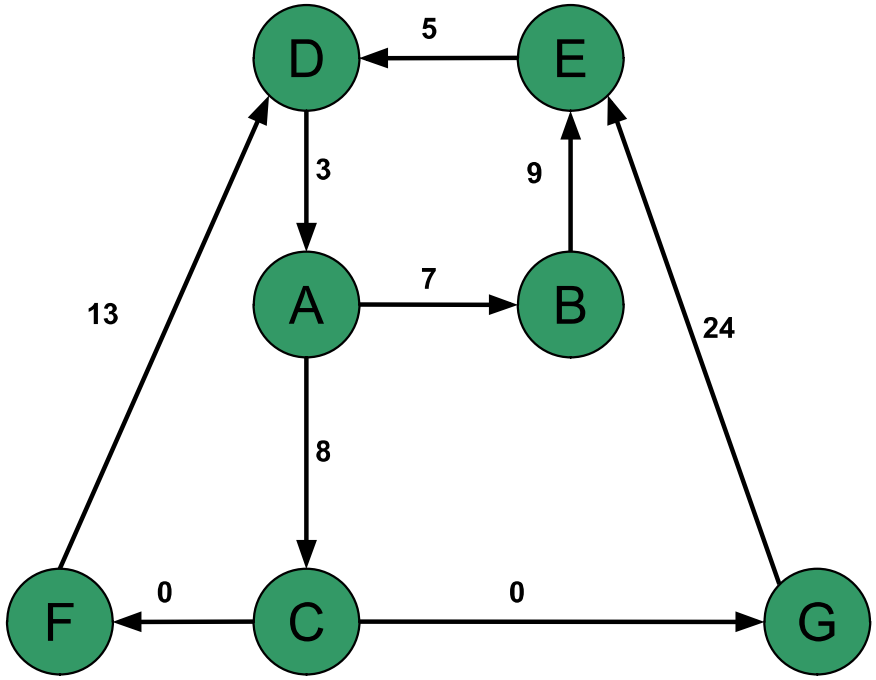


Fig 2c: Constructing a Majority Cotree

Constructing Majority Cotrees

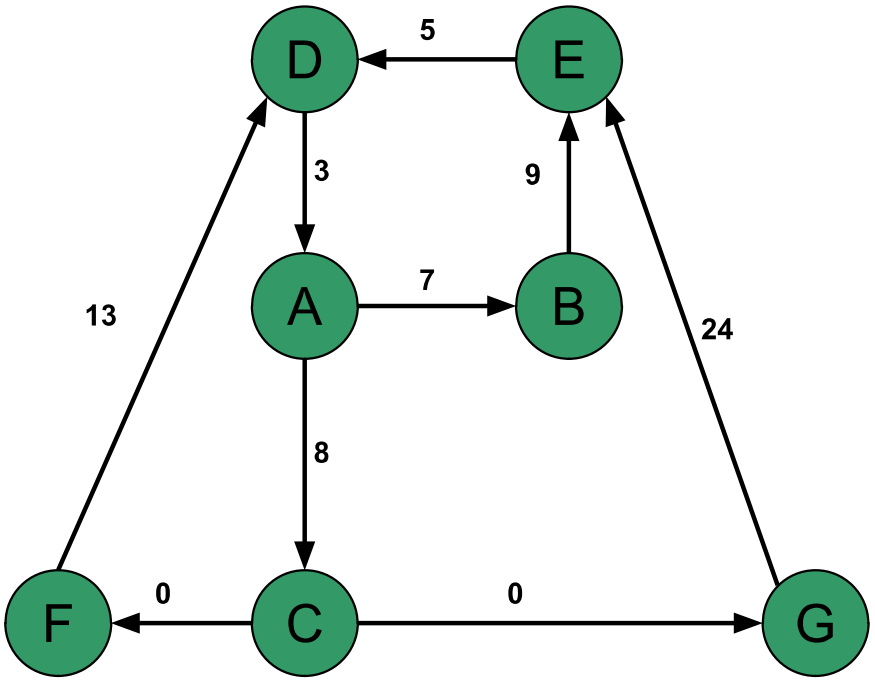


Fig 2c: Constructing a Majority Cotree(D)

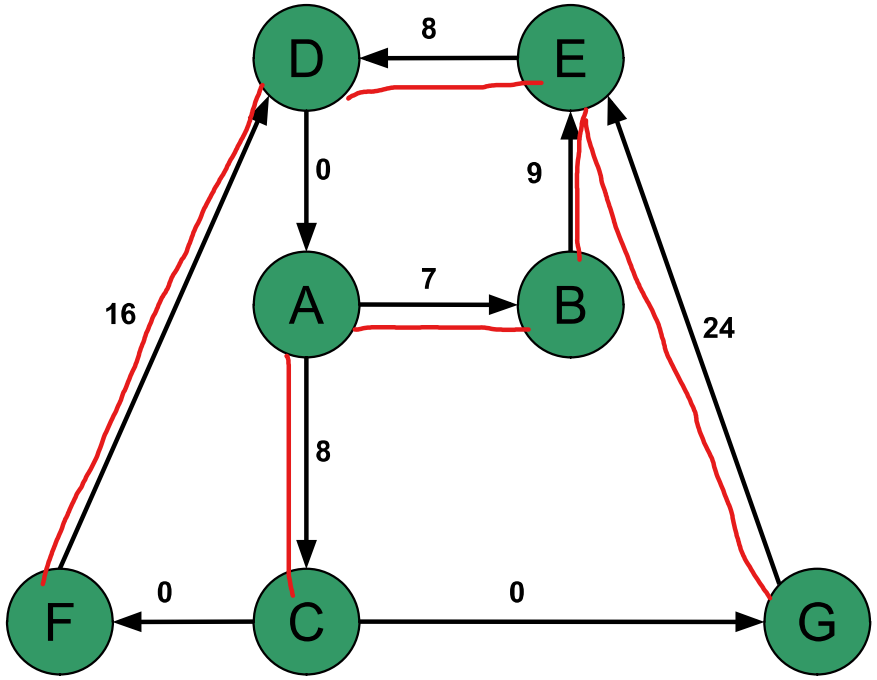


Fig 2d: Constructing a Majority Cotree

Constructing Majority Cotrees

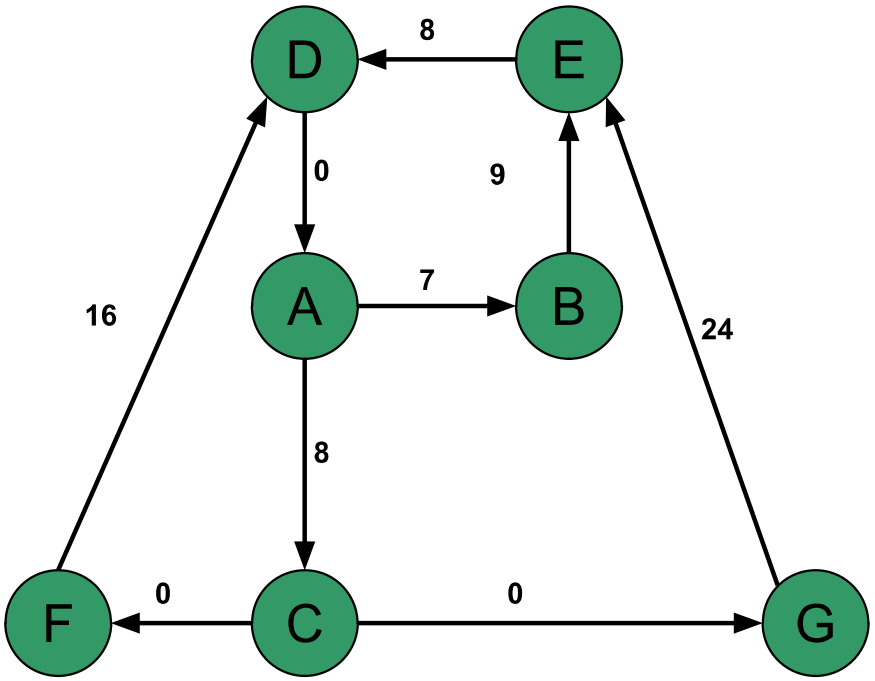


Fig 2d: Constructing a Majority Cotree(E)

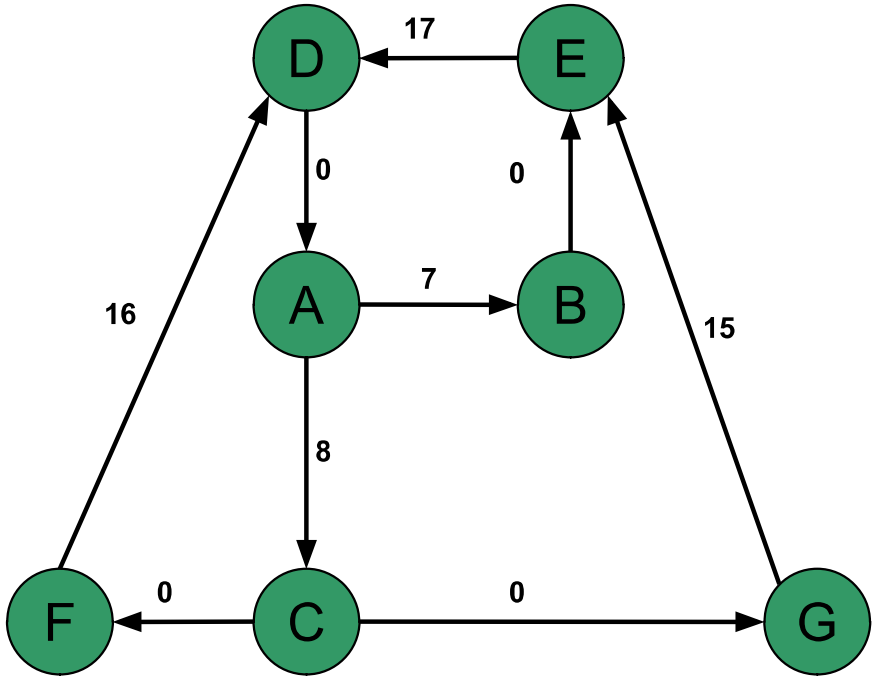


Fig 2e: Constructing a Majority Cotree

Constructing Majority Cotrees

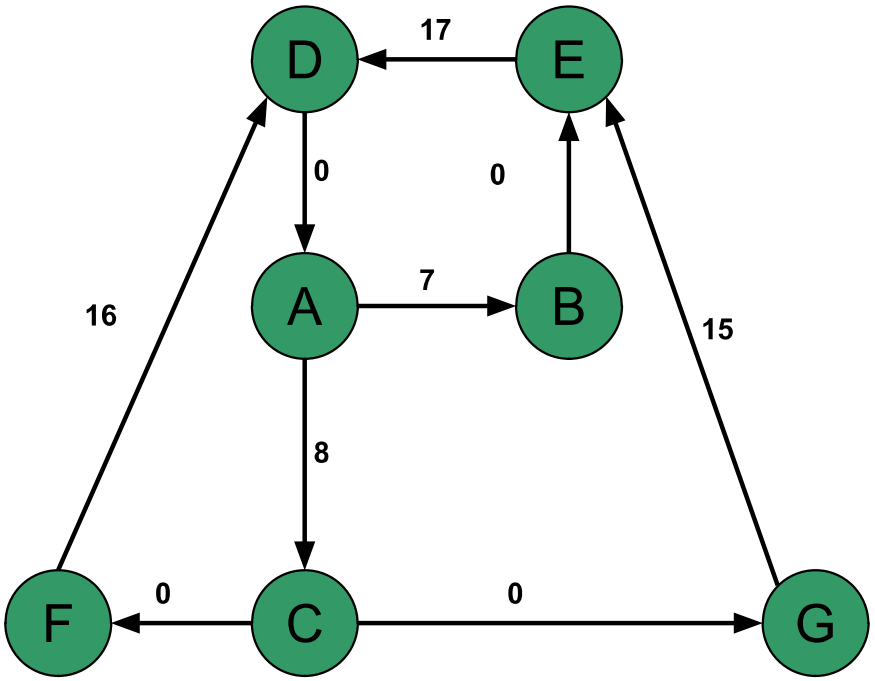


Fig 2e: Constructing a Majority Cotree(FCG)

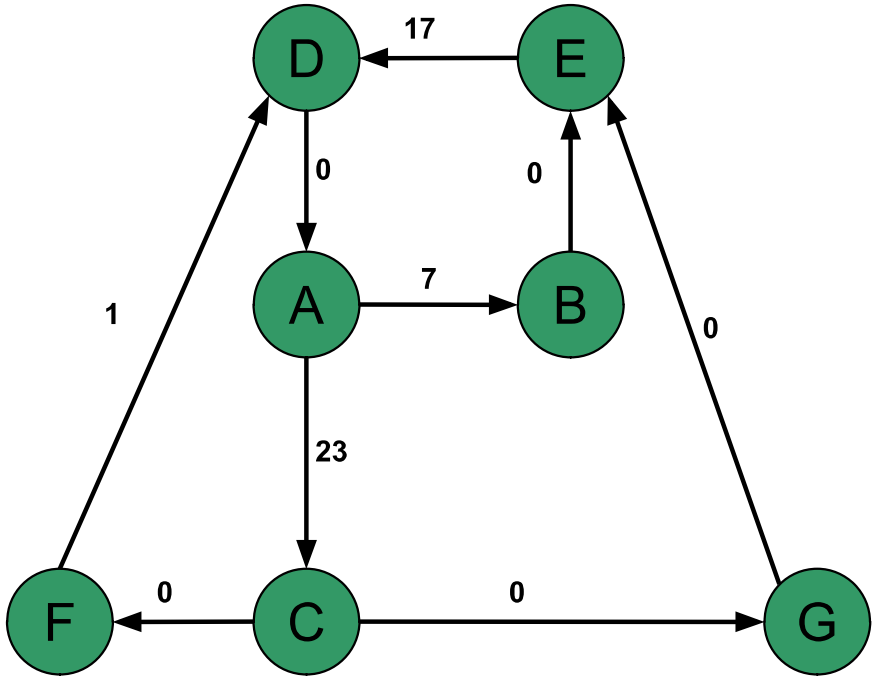


Fig 2f: Constructing a Majority Cotree

Constructing Majority Cotrees

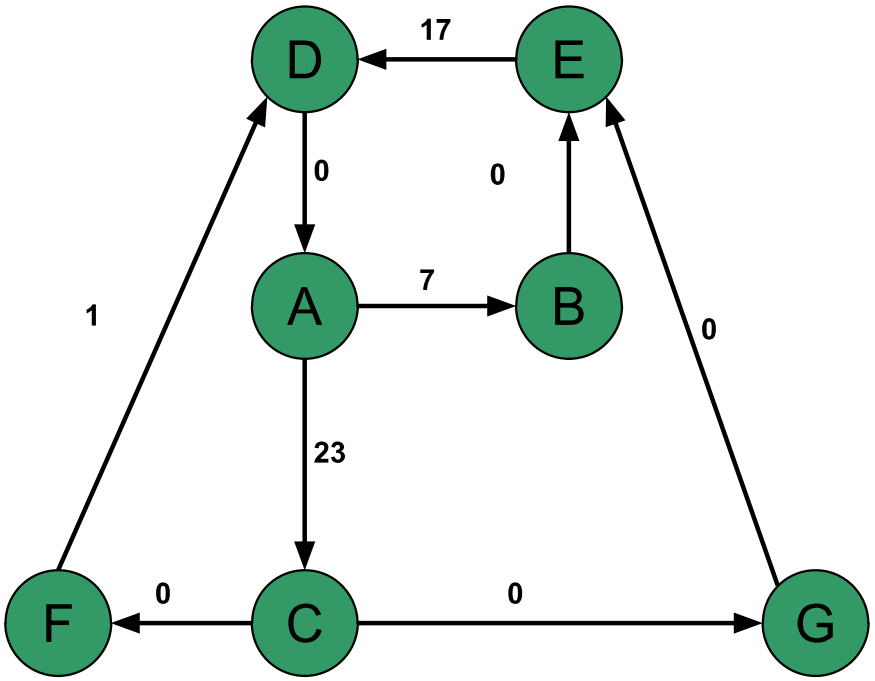


Fig 2f: Constructing a Majority Cotree(DA)

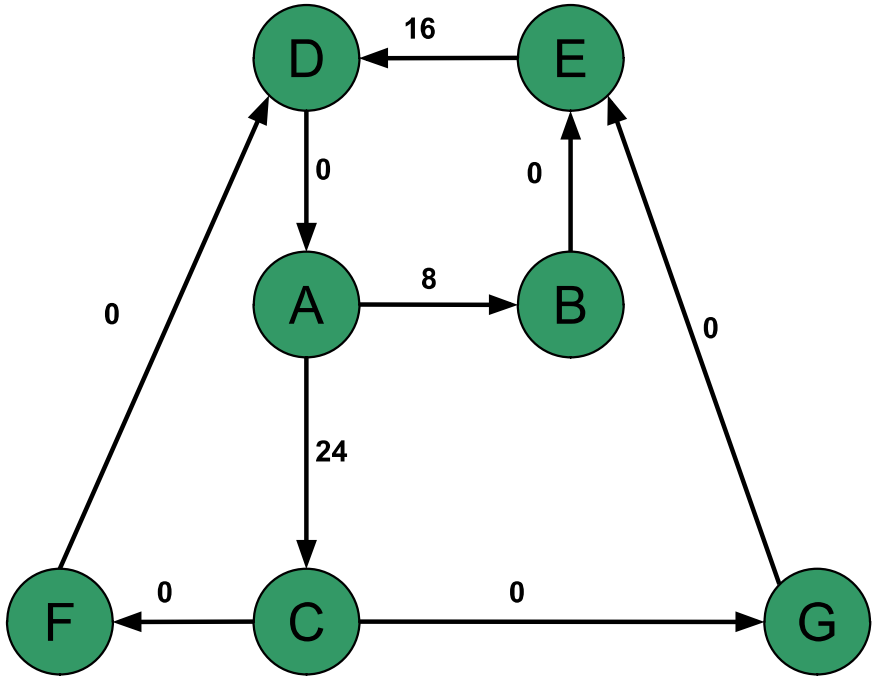


Fig 2g: Constructing a Majority Cotree

Constructing Majority Cotrees

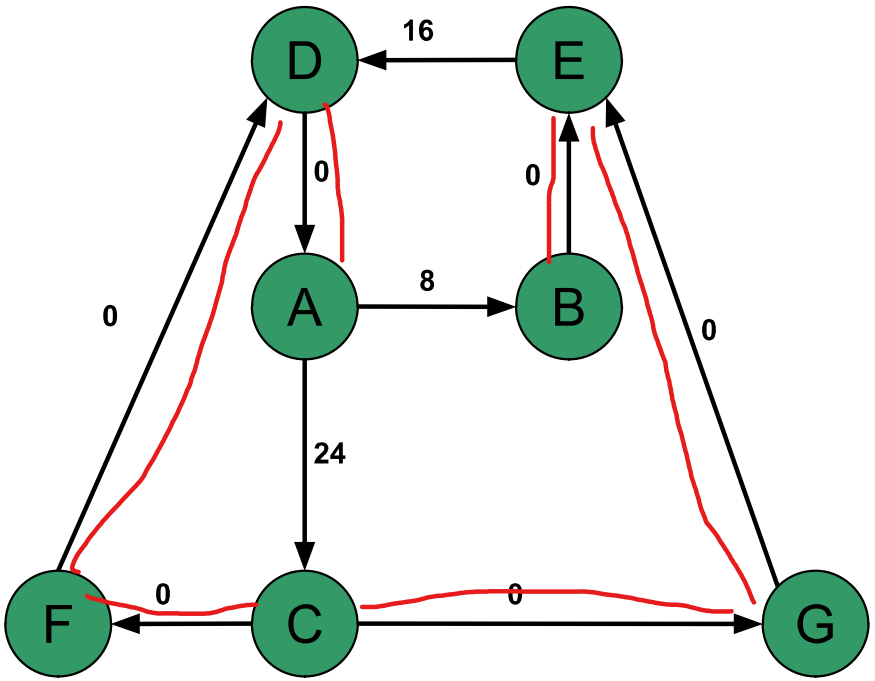


Fig 2g: Constructing a Majority Cotree

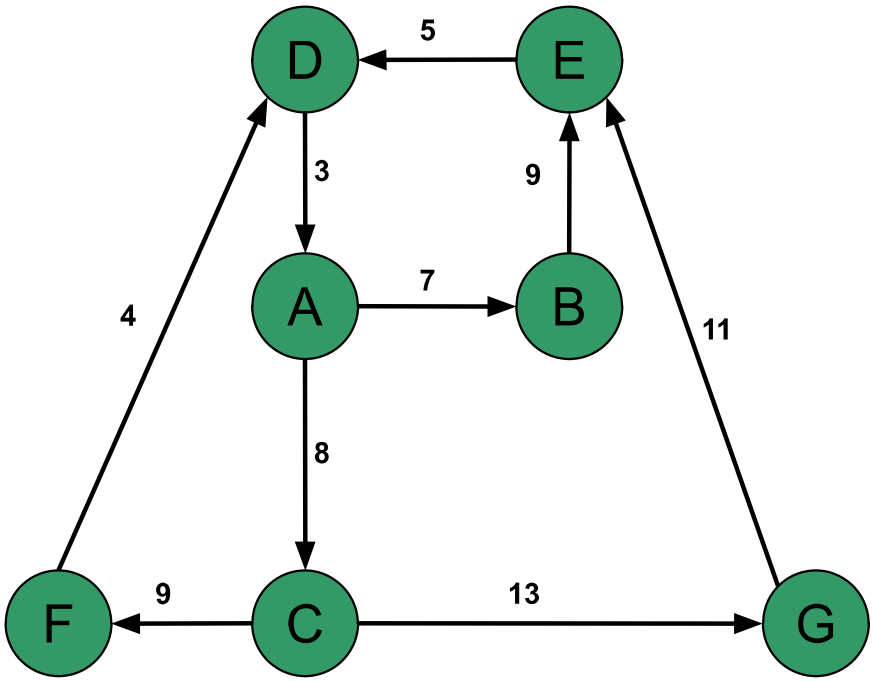


Fig 2a: Digraph with original weights

Constructing Majority Cotrees

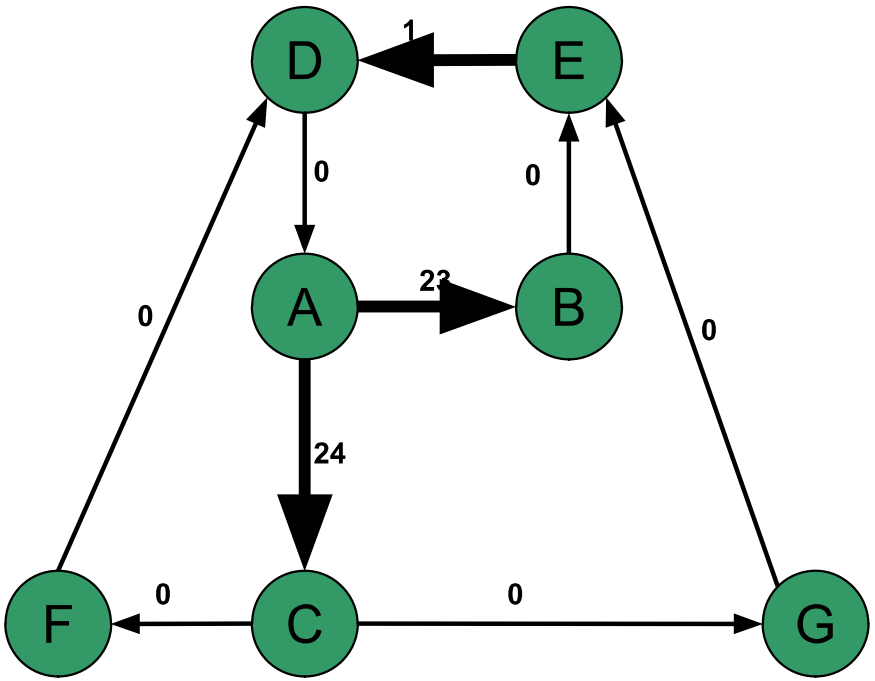
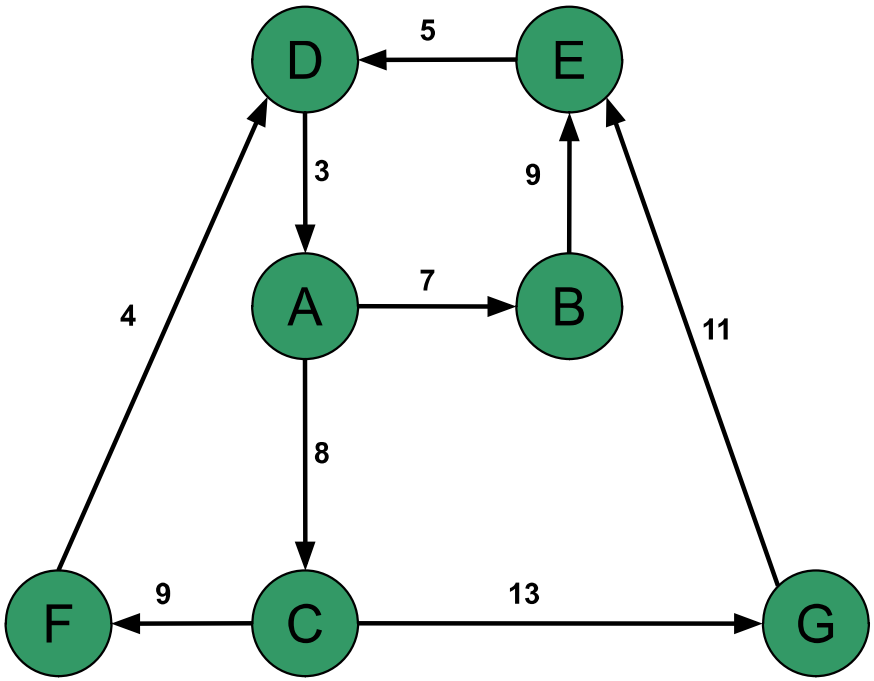


Fig 2f: A Majority Cotree



Digraph with original weights

A Linear Programming Proof of Theorem 1

- **Proof:** Let T be any arbitrary spanning tree of G, K_k be the fundamental cutset defined by edge .
- Now consider the following system of linear inequalities:

Qx=r, x >=0……..2.3
where, r is a column vector of length |V|-1 with components r_k, and X is a column vector of length |V|-1, components of which are indexed by the indices of edges of G. The system (2.3) is feasible since $x_{ij} = p_{ij}$, (i,j) ∈ E is a feasible solution. Since Q is of rank |V|-1, a basic feasible solution of (2.3) will have |V|-1 basic variables, which will correspond to edges of a spanning tree T* of G. Now the weights of the cut-sets, defined by the edges of T*, are the values of the basic variables; therefore, the value of each of the cut-sets, defined by the edges of T*, is non-negative. Hence T* is a majority spanning tree.

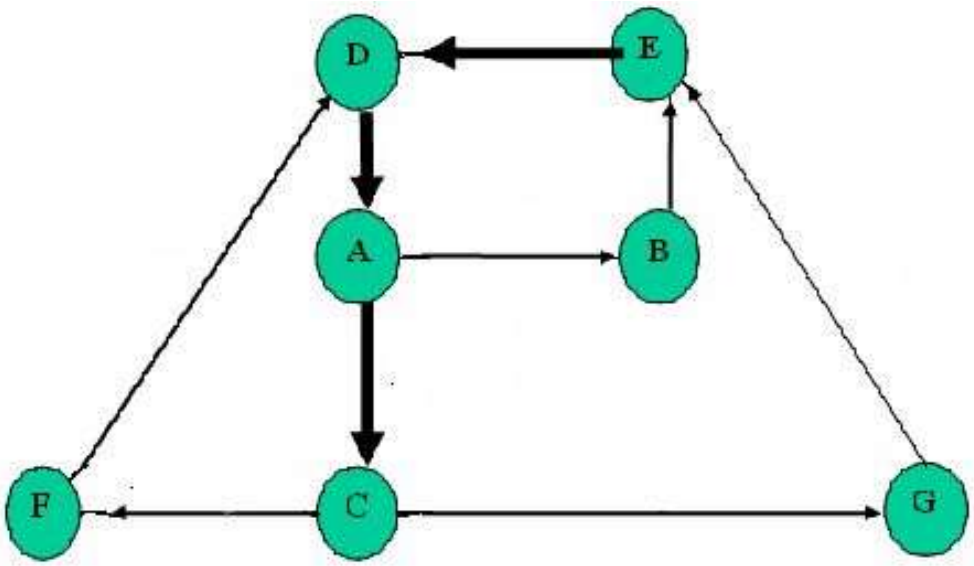
A Linear Programming Proof of Theorem 2

- **Proof:** Proof of this theorem is similar to that of Theorem 1. Only in this case we consider the following equations instead of (2.3):

$$Lw = s, w \geq 0 \qquad (2.4)$$

where, s is a column vector of length |E|-|V|+1 with components s_k as defined in (2.1), and w is a column vector of length |E|, the components of which are indexed by the edges of G. Basic feasible solution to (2.4) corresponds to a majority cotree.

A Majority Spanning Tree and Cotree Simultaneously



ED defines cycle {ED,DF,CF,CG,GE} with weight 5-4-9+13+11=16>=0
Fundamental cutset defined by arc AB has weight 7-3=4>=0

A Linear Programming Proof of Theorem 3

- **Proof:** Consider the following two optimization problems:

$$Max. \quad z_1 = -r^t w_T$$

$$Lw \le s, \quad w_T \ge 0 \qquad (2.5)$$

$$Min. \quad z_2 = -s^t y_{T'}$$

$$L_{T'}^t y_{T'} \le -r, \quad y_{T'} \ge 0 \qquad (2.6)$$

A Linear Programming Proof of Theorem 3(continue)

- Problem (2.6) is dual to problem (2.5). Since
- $$Q_{\bar{T}} = -L_T^t \qquad (2.7)$$

Any basic feasible solution to (2.6) corresponds to a majority cotree, and that of (2.5) corresponds to a majority spanning tree. Since optimal solution to either of (2.5) and (2.6) at the same time satisfy all the constraints of (2.5) and (2.6) that solution will correspond to majority spanning tree and a majority cotree. That is edge corresponding to basic variables of (2.5) will correspond to a majority spanning tree, and edges corresponding to non-basic variables of (2.5) will correspond to a majority cotree. This completes the proof.

Possible Applications

- **Minimum Connection Time Problem :**
  - In a transportation system it is not always possible to provide for all passengers a direct connection from origin to destination. Some passengers may have to use several connecting trips to reach their destination.
  - The connection time is the time a passenger has to wait between connecting trips.
  - The problem that addresses how to construct a timetable for the transportation system that minimizes the total weighted connection time will be referred to as *Minimum Connection Time (McT) Problem*

McT Problem

- Formulation in a Digraph
- Node corresponds to trip, and edge corresponds to a connection.
- t_i, x_i, p_{ij} correspond to trip time, trip starting time and # of passengers taking trip j after completing i, and
- w_{ij} corresponds to waiting time in connection.

Theorem For MCT Problem

- Let Let *p* be the weight of edges of *G(V,E)*. Then
- **Theorem 4** For the optimal solution to the MCT problem, the set of edges with $w_{ij} = 0$ contains a majority spanning tree.

Mathematical Model

$$w_{ij} = (x_j - x_i - t_i) \bmod \tau \tag{3.1}$$

with

$$0 \le w_{ij} < \tau \tag{3.2}$$

The total weighted connection time in the system is equal to

$$z = \sum_{(i,j) \in E} p_{ij} w_{ij} \tag{3.3}$$

Complexity Issues of MCT Problem

INSTANCE: Given an arbitrary directed graph G=(V,E) with non-negative integer p_{ij} (number of passengers waiting for connecting trip j), $(i,j) \in E$ non-negative integers t_i (trip time), $i \in V$, and positive integers τ and B.

QUESTION: Do there exist integers $x_i, i \in V$ such that for $w_{ij} = (x_j - x_i - t_i) \bmod \tau$, $\sum_{(i,j) \in E} p_{ij} w_{ij} \leq B$?

Theorem 5 MCT Problem is NP-Complete.

Proof: It is easy to show that MCT belongs to the class NP since any yes answer to it can be verified in polynomial time by calculating the left hand side of the inequality. We prove this theorem by transforming arbitrary instance of MAX CUT problem to an instance of MCT problem. Given an arbitrary instance of MAX-CUT on $H = (V, E_1)$ weight p_{ij} on its edges, and a bound S, we construct an instance of MCT problem in the following way. Let G have the same vertices as H and construct E by giving arbitrary orientation to E_1, and let edges receive the same weights as in H. Now set $t_i = 1, \forall i \in V, r = 2$ and $B = \sum_{(ij)\in E} p_{ij} - S$. $V_1 \cup V_2 = V$ Now MAX CUT asks the question: is there a bipartite graph with a total weight S or more. If the answer is yes then let the bipartitions are V_1 and V_2 with $V_1 \cup V_2 = V, V_1 \cup V_2 = \varnothing$. Now set $x_i = 0, \forall i \in V_1$ and $x_i = 1, \forall i \in V_2$. If there is a MAX CUT with weight S or more then the

NP-completeness Proof (Continued)

weight of all the edges connecting vertices both in V_1 or both in V_2 is less than or equal to $\sum_{(i,j)\in E} p_{ij} - S = B$. Then connections corresponding to edges running from one partition to another will have no waiting time since for them $x_j - x_i - t_i = 1 - 0 - 1 = 0, i \in V_1, j \in V_2$. Whereas for each edge with end vertices $i, j \in V_1$ or, $i, j \in V_2, (x_j - x_i - t_i) \bmod \tau = (x_j - x_i - 1) \bmod 2 = 1$ since the values of x_i and x_j are the same. On the other hand assume that there is a "yes" answer to MCT problem. Then waiting time in each edge can be either 0 in which case for end vertices i, j, $x_i \neq x_j$ or can be 1 in which case for end vertices i, j, $x_i = x_j$ and waiting time will be 1 for each such connection. Then putting vertices $V_1 = \{i \mid x_i = 0\}$ and $V_2 = \{j \mid x_j = 1\}$ will give us a bipartite subgraph, where we have deleted only the edges in the same partition that have the total weight of S or less. Hence total weight of edges in the bipartite subgraph will be at least $\sum_{(i,j)\in E} p_{ij} - S = B$. This proves that both the MAX CUT and MCT problems share the same complexity status. Since MAX CUT is NP-Complete[4], so must be MCT problem.

Possible Applications (cont.)

- **Ranking Problem:**
  - Consider the problem of ranking players of a round-robin tournament by minimizing number of upsets, that is number of matches in which lowly ranked players have defeated highly ranked players.
  - The results of such a tournament can be expressed in a digraph known as *tournament digraph*.
  - In a tournament digraph players correspond to vertices and each edge corresponds to the result of a match in which case edge starts at the vertex corresponding to the winning player, and the defeated player corresponds to the head of the edge.

Theorem For Ranking Problem

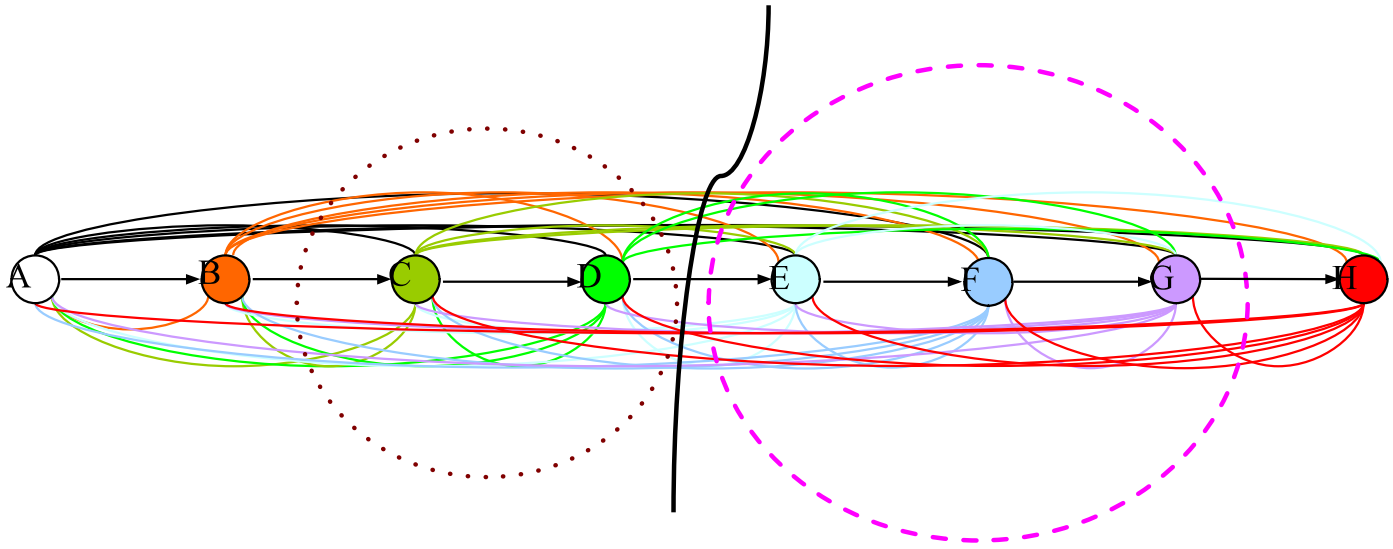
- Let *R* be a ranking of players,
  - $V_R = V$ and
  - $E_R = \{(i,j)|$ rank of player corresponding to vertex j is immediately below that of player corresponding to vertex i$\}$.
  - It is obvious that $G_R=(V_R,E_R)$ is a spanning tree of G, more accurately $G_R=(V_R,E_R)$ is a Hamiltonian semipath.
- **Theorem 6:** *Let R be any optimal ranking of a tournament represented by G=(V,E). Then* $G_R=(V_R,E_R)$ *is an MST of G.*

Theorem For Ranking Problem (cont.)

- Let,
  - G^{ij}® be the subgraph of G induced by the set of vertices corresponding to players ranked from i to j, and
  - $G^{ij}R$ be the subgraph of G^{ij}® having the same set of vertices, and only those edges that correspond to the results of matches between consecutively ranked players. Then

- **Theorem 7:** For any optimal ranking R and $1 \leq i \leq j \leq n$, $G^{ij}R$ must be an MST of G^{ij}®.

MST Algorithmic Concept for Round Robin Tournament Ranking



CE, DE: Forward
FC, GC, FD, GD: Backward

A one-way road network equilibrium problem

- Assume that there has been an equilibrium problem of traffic flow in different directions. Let G=(V,E) show requirement for equilibrium with edge weights representing cost of constructing roads for a unit traffic. Then we are looking for a majority spanning tree that minimizes appropriate cost function.

Settling Multiple Debts Efficiently

- Consider a set of people among whom some borrowing have taken place. Now these debts have to be settled through money exchanges only among those who have lending-borrowing relations.
- We can construct a directed graph with edge weights equal to the amount of money borrowed-borrower is the tail and lender is the head.

Settling Multiple Debts Efficiently(contd)

- Now we can easily construct a majority spanning tree that will show how this can be settled. However, to reduce the number of exchanges we can simply identify vertices (people) that represent net borrowers and that represent net lenders. These amounts can be easily distributed reducing number of exchanges using the majority spanning tree.

Conclusion

- We have discovered a new class of spanning trees, termed as *majority spanning trees,and* majority cotrees in directed graphs with non-negative weights on edges.
- We have proved co-existence of majority spanning trees and majority cotrees in a directed graph.

Conclusion(contd)

We have shown their applications to

- Minimum Connection Time Problem
- Optimal Ranking of a Round Robin Tournament
- in complementing a one-way road network for equilibrium
- in settling multiple debts.

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End of lecture

Thank you very much for your passionate attention.